Large DN pipe wall thickness measurement machine development

Saint-Gobain Pipeline (Maanshan) Co., ltd

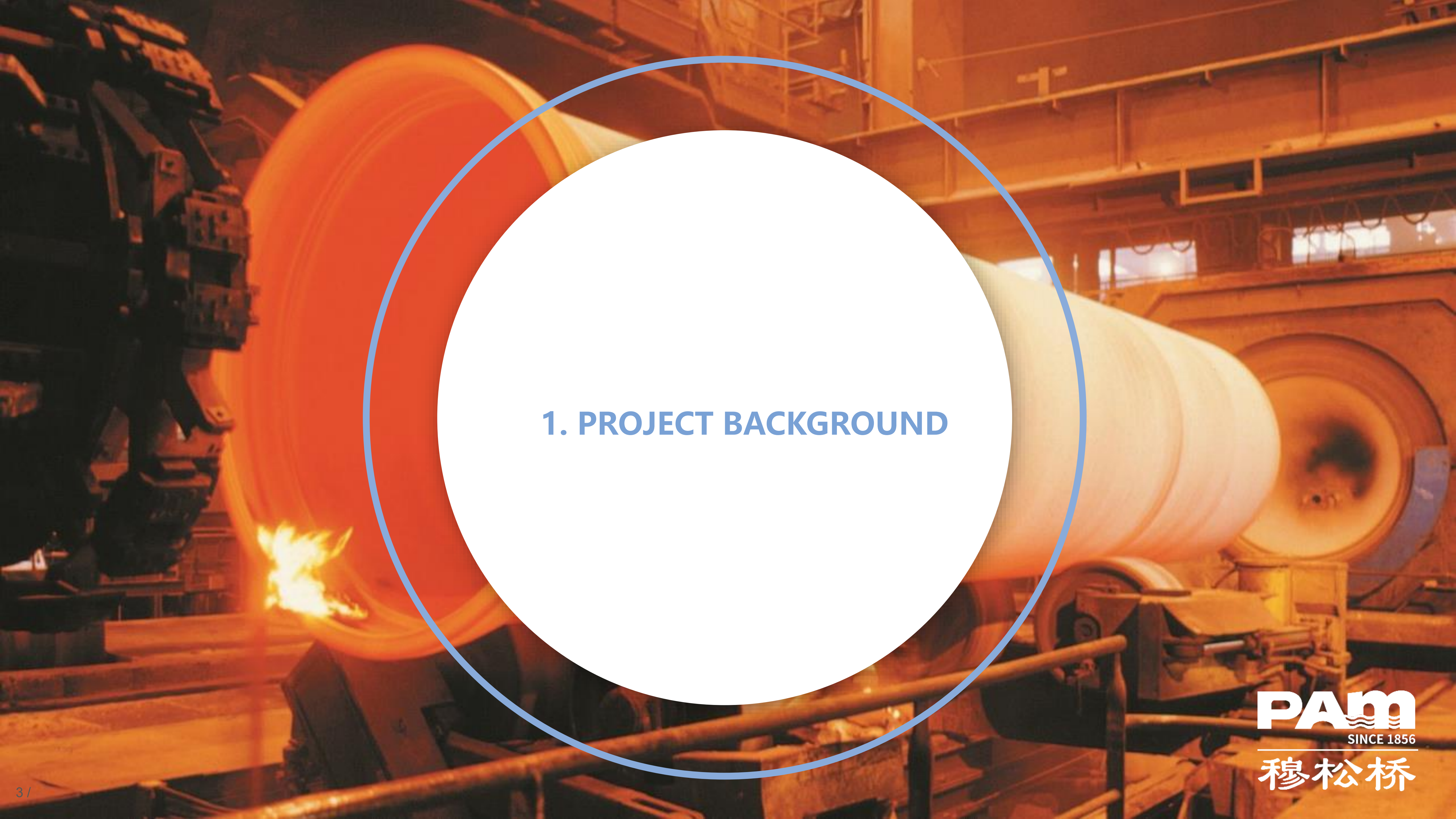
CHEN Bohan
Intern student from UC Davis

SINCE 1856
穆松桥

目录

CONTENT

PAM SINCE 1856 穆松桥



1. PROJECT BACKGROUND

PROJECT BACKGROUND

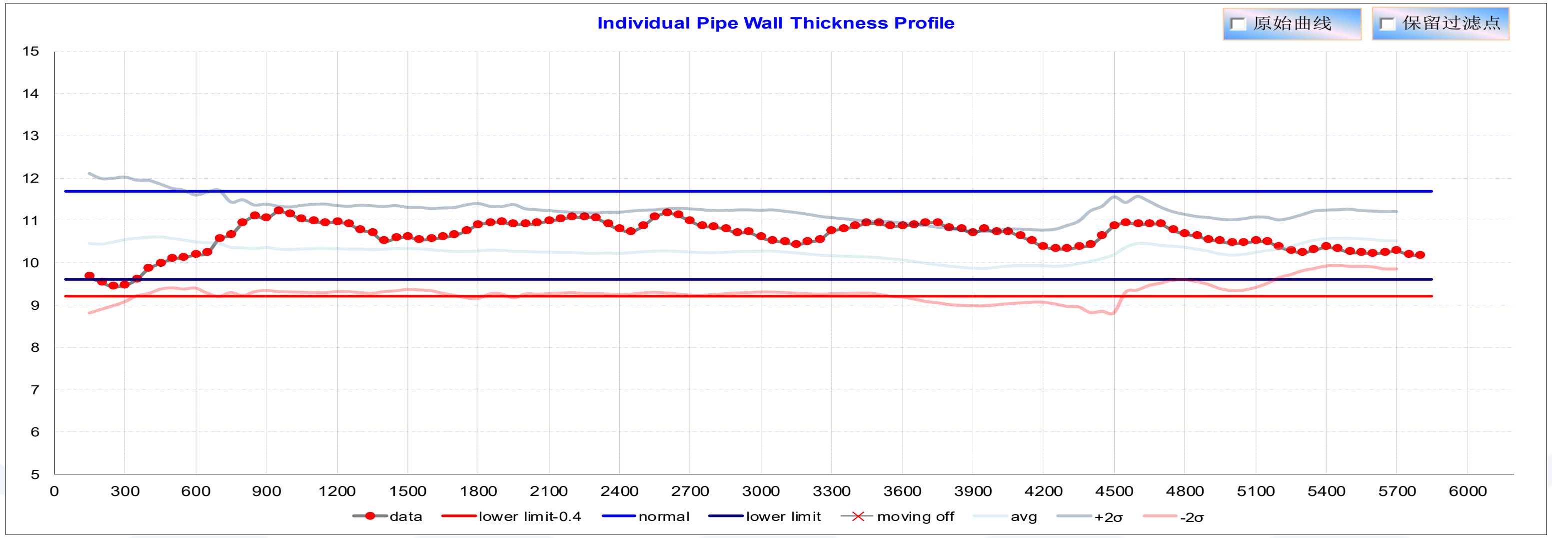
The ultrasonic wall thickness measurement machine

In pipe production, wall thickness measurement is to

- Make sure that the wall thickness of pipe can meet the requirement of minimum thickness defined in the standard

.

- Have loop to guide the casting operators to cast the pipe with as min thickness as possible to reduce the cost.



PROJECT BACKGROUND

The ultrasonic wall thickness measurement machine

The PAM is the only company that use their self-developed ultrasonic wall thickness measurement machine to detect the wall thickness of the pipe. But this technology

- Can only be used for measure the pipe from DN80-DN1000.
- Will have very bad signals for large DN, which means DN1200 and above.
- The casting process of DN1200 and above is totally different from DN80-DN1000, which will cause the inner wall surface very rough, and uneven, so it is difficult for ultrasonic solution to have good signals.

→ So it is important and urgent to develop a machine to measure the wall thickness of DN1200 and above !!

PROJECT BACKGROUND

Internship company

- Saint-Gobain Pipeline (MAS) Co., Ltd
- Located in Ma'Anshan city, Anhui Province, China
- Belongs to PAM France

Group member

- CHEN Bohan: internship student from UC Davis
  - ☞ Collect data and use excel to do analysis
  - ☞ Calculate to see if feasible to find cost optimized pipe weight
  - ☞ Design the software blueprint
  - ☞ Communicate with SG Shanghai R&D center for software development by using blueprint
- SUN Xisen: R&D engineer in Saint-Gobain Pipeline (Maanshan) Co., ltd
  - ☞ As a tutor to guide / support what CHEN Bohan is doing
- WANG Zhiyi: Software development Engineer in Saint-Gobain Shanghai Research Center (SGRS)
  - ☞ Develop the software according software blueprint
  - ☞ Frequent communication during members is necessary

2. Technical feasibility study

TECHNICAL FEASIBILITY STUDY

PAM SINCE 1856 穆松桥

Basic requirements of wall thickness measurement machine

- Capable of measuring the wall thickness of pipes with precision 0.1mm
- Automatic data recording, storage and export from the backend
- Meet production safety requirements
- Capable of resisting vibration, high temperature, noise during the production
- Easy to maintain, operate and running cost is as cheap as possible

TECHNICAL FEASIBILITY STUDY

Technical solution investigation

- ***Literature:***

  - The French PAM research center has done research on laser wall thickness measurement machine and electromagnetic ultrasonic wall thickness measurement machine.

  - After reading these articles published by the PAM Research Center, it was found that these two wall thickness measurement techniques are not being used in practical application.

- ***Patents:***

  - Xin Xing cast pipe company received a patent related to laser thickness measurement

  - However, after research, it was found that the patent has not been applied in practical applications.

  - The patent is only a basic experiment on laser thickness measurement in the laboratory and the technology has not been researched to the level that can be applied in the factory.

TECHNICAL FEASIBILITY STUDY

Technical solution investigation

- ***Famous suppliers：***

- Keyence, a famous company that makes sensors. After consulting with their technical staff about several different sensors, it was found that electromagnetic ultrasonic sensors could not be used for actual wall thickness inspection. Here are several reasons：

    - Electromagnetic ultrasonic thickness measurement still belongs to the category of ultrasonic, so it will be more affected when measuring large-diameter pipes.

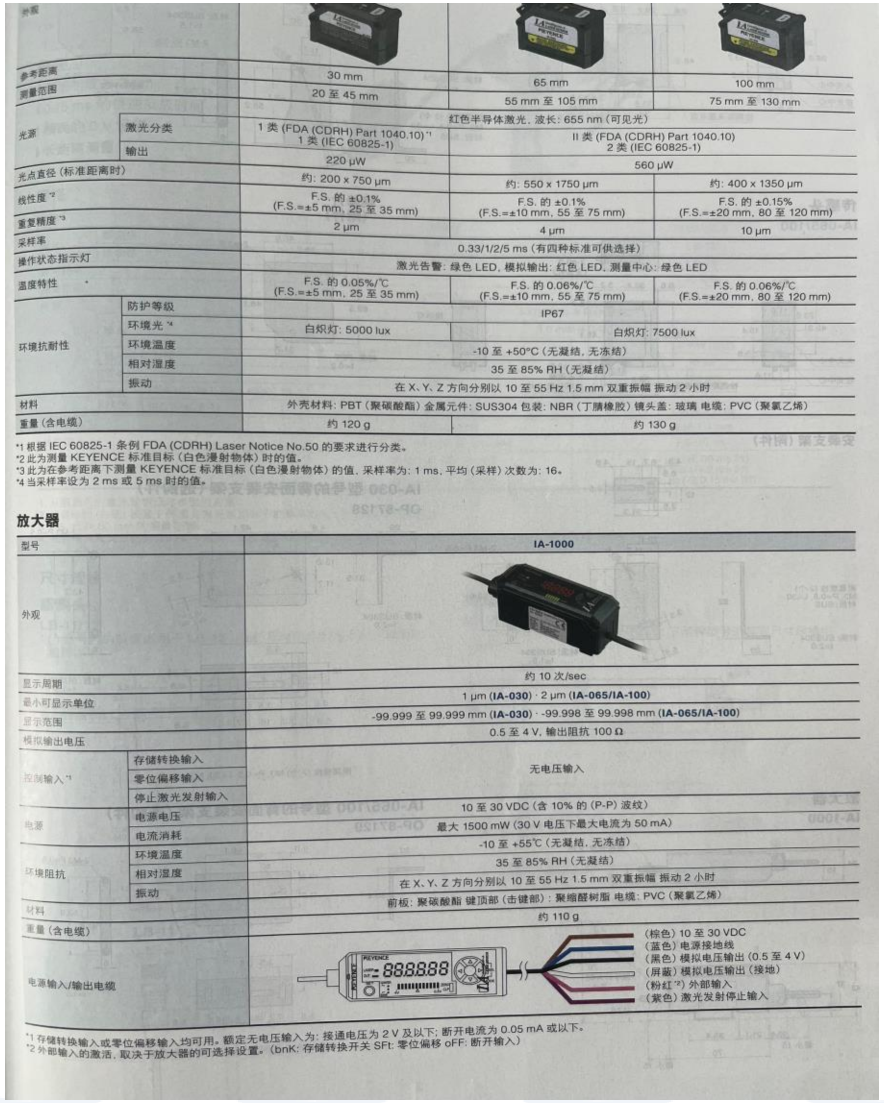
    - Electromagnetic ultrasound technology is not very mature and there is a limited number of brand that can offer this equipment, leaving little room for choice.

→ After analysis of all potential technical solutions, the laser thickness measurement machine is highly feasible.

PAM SINCE 1856 穆松桥

Sensor brand and sensor selection

- We choose the Keyence brand to pick the right sensors.
- CMOS laser displacement sensor IL-100
- CMOS analog laser sensors IA100
- CCD laser displacement sensor LK-G150/155



外观				
参考距离		30 mm	65 mm	100 mm
测量范围		20 至 45 mm	55 mm 至 105 mm	75 mm 至 130 mm
光源		红色半导体激光，波长：655 nm（可见光）		
	激光分类	1 类 (FDA (CDRH) Part 1040.10)*1 1 类 (IEC 60825-1)	II 类 (FDA (CDRH) Part 1040.10) 2 类 (IEC 60825-1)	
	输出	220 µW	560 µW	
光点直径（标准距离时）		约：200 x 750 µm	约：550 x 1750 µm	约：400 x 1350 µm
线性度*2		F.S. 的 ±0.1% (F.S.=±5 mm，25 至 35 mm)	F.S. 的 ±0.1% (F.S.=±10 mm，55 至 75 mm)	F.S. 的 ±0.15% (F.S.=±20 mm，80 至 120 mm)
重复精度*3		2 µm	4 µm	10 µm
采样率		0.33/1/2/5 ms（有四种标准可供选择）		
操作状态指示灯		激光告警：绿色 LED，模拟输出：红色 LED，测量中心：绿色 LED		
温度特性		F.S. 的 0.05%/℃ (F.S.=±5 mm，25 至 35 mm)	F.S. 的 0.06%/℃ (F.S.=±10 mm，55 至 75 mm)	F.S. 的 0.06%/℃ (F.S.=±20 mm，80 至 120 mm)
环境抗耐性	防护等级	IP67		
	环境光*4	白炽灯：5000 lux	白炽灯：7500 lux	
	环境温度	-10 至 +50°C（无凝结，无冻结）		
	相对湿度	35 至 85% RH（无凝结）		
	振动	在 X、Y、Z 方向分别以 10 至 55 Hz 1.5 mm 双重振幅 振动 2 小时		
材料		外壳材料：PBT（聚碳酸酯）金属元件：SUS304 包装：NBR（丁腈橡胶）镜头盖：玻璃 电缆：PVC（聚氯乙烯）		
重量（含电缆）		约 120 g	约 130 g	

*1 根据 IEC 60825-1 条例 FDA (CDRH) Laser Notice No.50 的要求进行分类。
*2 此为测量 KEYENCE 标准目标（白色漫射物体）时的值。
*3 此为在参考距离下测量 KEYENCE 标准目标（白色漫射物体）的值，采样率为：1 ms，平均（采样）次数为：16。
*4 当采样率设为 2 ms 或 5 ms 时的值。

放大器

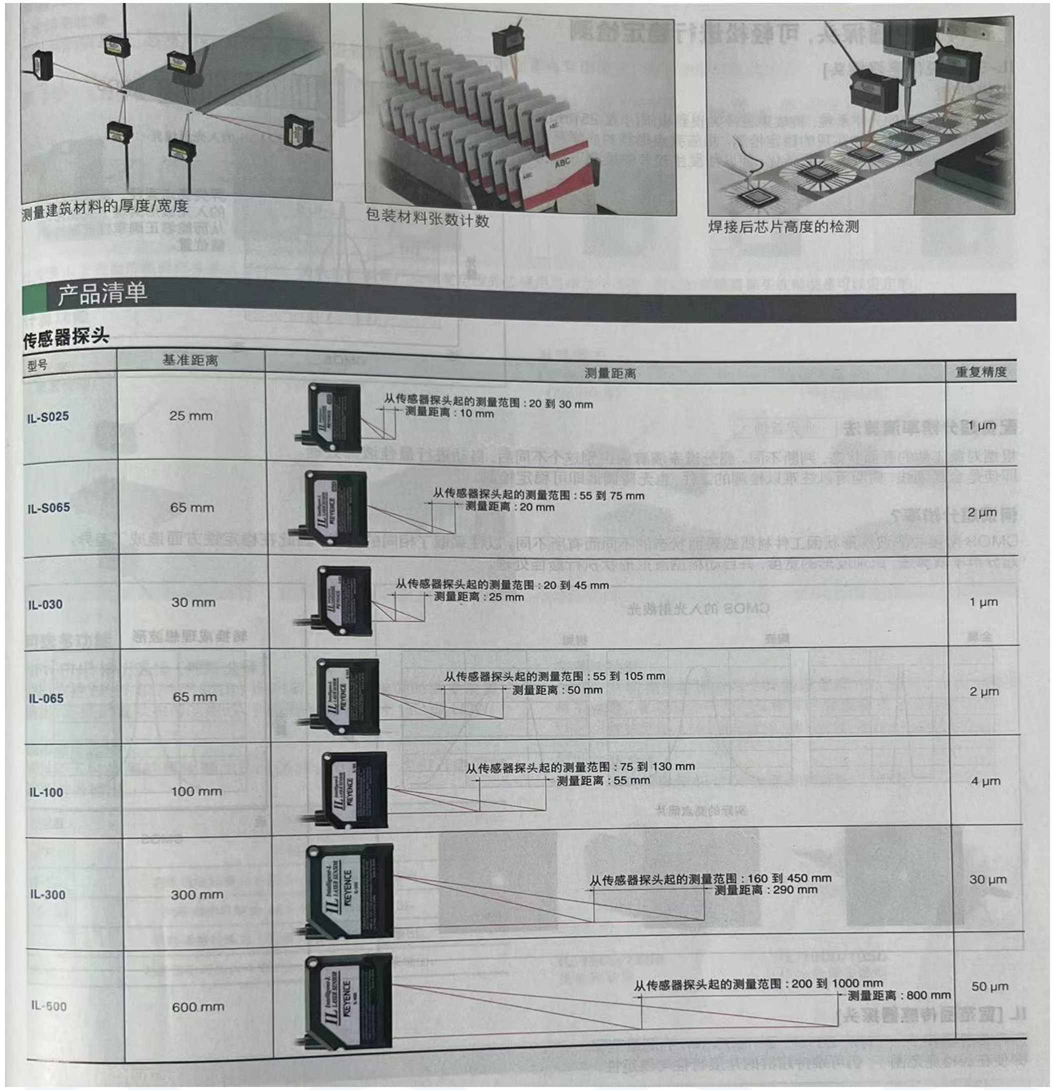
型号		IA-1000
外观		
显示周期		约 10 次/sec
最小可显示单位		1 µm (**IA-030**) · 2 µm (**IA-065/IA-100**)
显示范围		-99.999 至 99.999 mm (**IA-030**) · -99.998 至 99.998 mm (**IA-065/IA-100**)
模拟输出电压		0.5 至 4 V，输出阻抗 100 Ω
控制输入*1	存储转换输入	无电压输入
	零位偏移输入	
	停止激光发射输入	
电源	电源电压	10 至 30 VDC（含 10% 的 (P-P) 波纹）
	电流消耗	最大 1500 mW（30 V 电压下最大电流为 50 mA）
环境阻抗	环境温度	-10 至 +55℃（无凝结，无冻结）
	相对湿度	35 至 85% RH（无凝结）
	振动	在 X、Y、Z 方向分别以 10 至 55 Hz 1.5 mm 双重振幅 振动 2 小时
材料		前板：聚碳酸酯 键顶部（击键部）：聚缩醛树脂 电缆：PVC（聚氯乙烯）
重量（含电缆）		约 110 g
电源输入/输出电缆		（棕色）10 至 30 VDC （蓝色）电源接地线 （黑色）模拟电压输出（0.5 至 4 V） （屏蔽）模拟电压输出（接地） （粉红*2）外部输入 （紫色）激光发射停止输入

*1 存储转换输入或零位偏移输入均可用。额定无电压输入为：接通电压为 2 V 及以下；断开电流为 0.05 mA 或以下。
*2 外部输入的激活，取决于放大器的可选择设置。（bnK：存储转换开关 SFt：零位偏移 oFF：断开输入）

TECHNICAL FEASIBILITY STUDY

Technical data of different sensors

- CMOS laser displacement sensor IL-100
  - ➢ Reference distance: 100mm
  - ➢ Measuring distance: 55mm
  - ➢ Repearability accuracy: 4 micron
- CMOS analog laser sensors IA100
  - ➢ No wall thickness test cases
- CCD laser displacement sensor LK-G150/155
  - ➢ Measuring distance: 110-190mm



测量建筑材料的厚度/宽度

包装材料张数计数

焊接后芯片高度的检测

产品清单

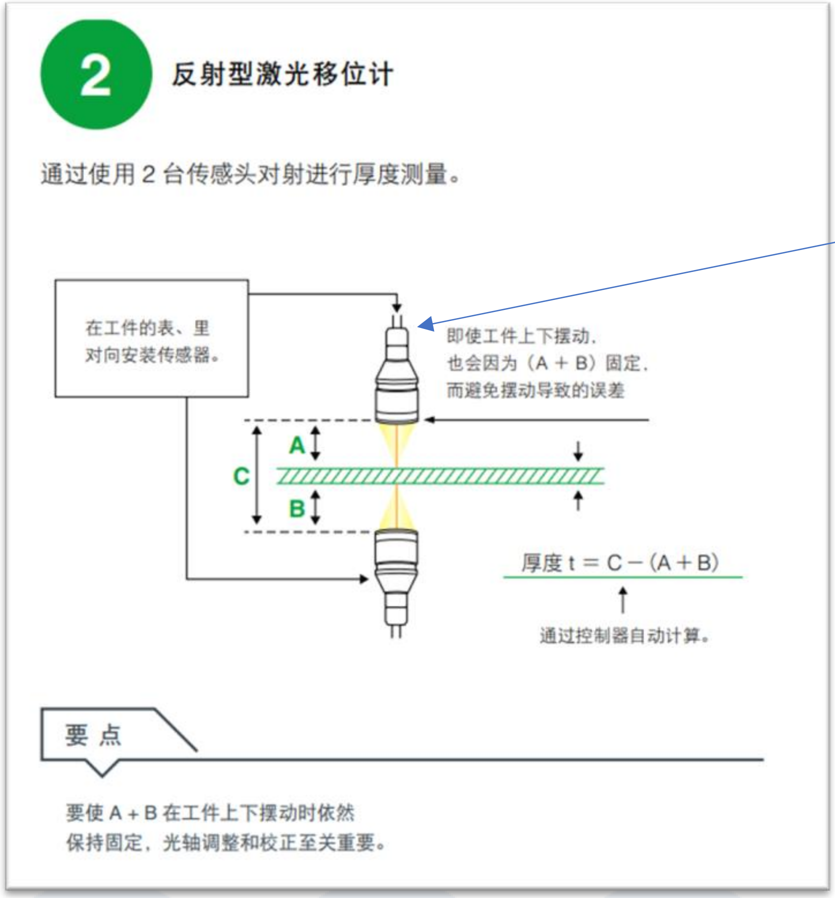
传感器探头

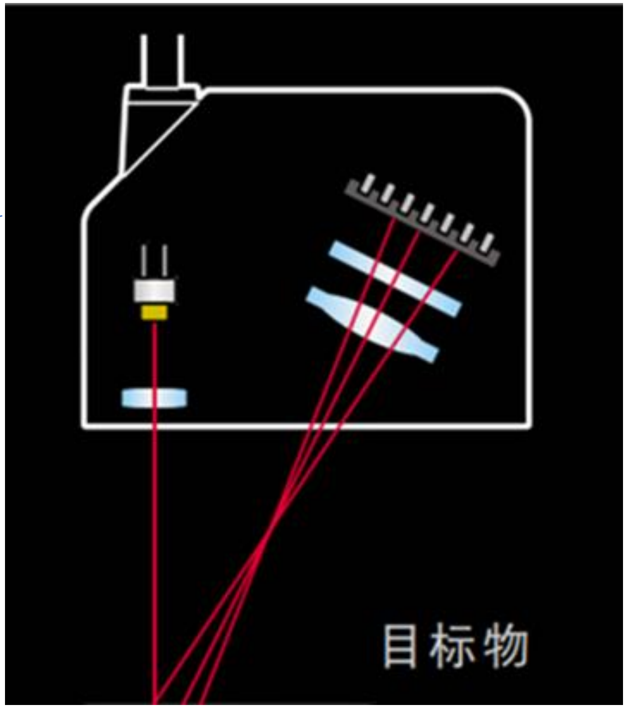
型号	基准距离	测量距离	重复精度
IL-S025	25 mm	从传感器探头起的测量范围 : 20 到 30 mm 测量距离 : 10 mm	1 μm
IL-S065	65 mm	从传感器探头起的测量范围 : 55 到 75 mm 测量距离 : 20 mm	2 μm
IL-030	30 mm	从传感器探头起的测量范围 : 20 到 45 mm 测量距离 : 25 mm	1 μm
IL-065	65 mm	从传感器探头起的测量范围 : 55 到 105 mm 测量距离 : 50 mm	2 μm
IL-100	100 mm	从传感器探头起的测量范围 : 75 到 130 mm 测量距离 : 55 mm	4 μm
IL-300	300 mm	从传感器探头起的测量范围 : 160 到 450 mm 测量距离 : 290 mm	30 μm
IL-600	600 mm	从传感器探头起的测量范围 : 200 到 1000 mm 测量距离 : 800 mm	50 μm

TECHNICAL FEASIBILITY STUDY

SINCE 1856
穆松桥

Schematic diagram of the sensor principle

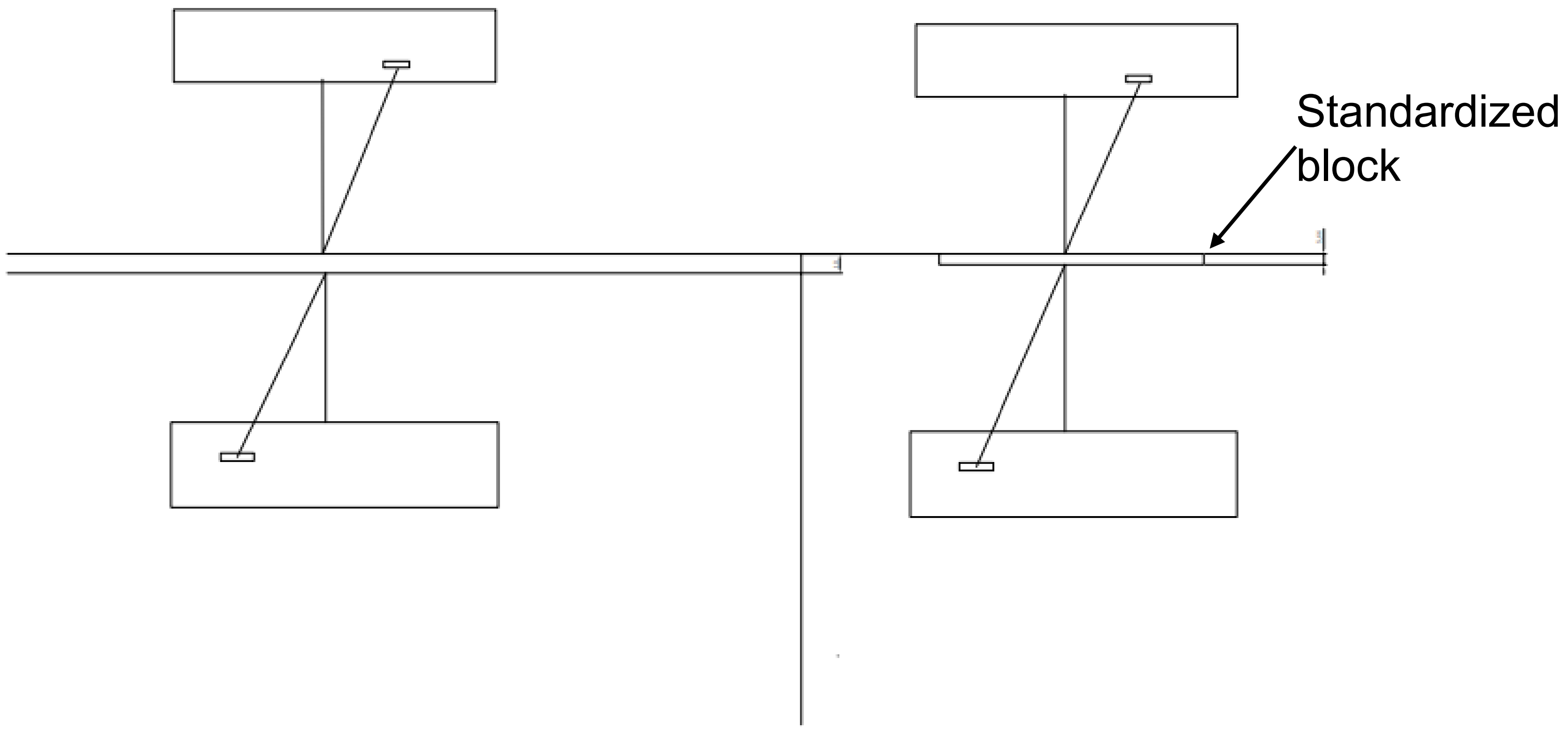




- The principle is to emit laser light in the object to be measured to produce diffuse reflection.
- The laser light in the diffuse reflection is received by the receiver, so as to measure the distance A or distance B.

PAM SINCE 1856 穆松桥

Schematic diagram of the sensor principle



Formula:

- T+ upward displacement + lower displacement

Comments:

- Positive displacement near the gauge, negative displacement away from the gauge.

→ After analyzing the principle and theoretical experiments, it was concluded that laser thickness measurement machine is feasible. Then did some small practical experiments to select suitable models.

Wall thickness measurement sensor final selection

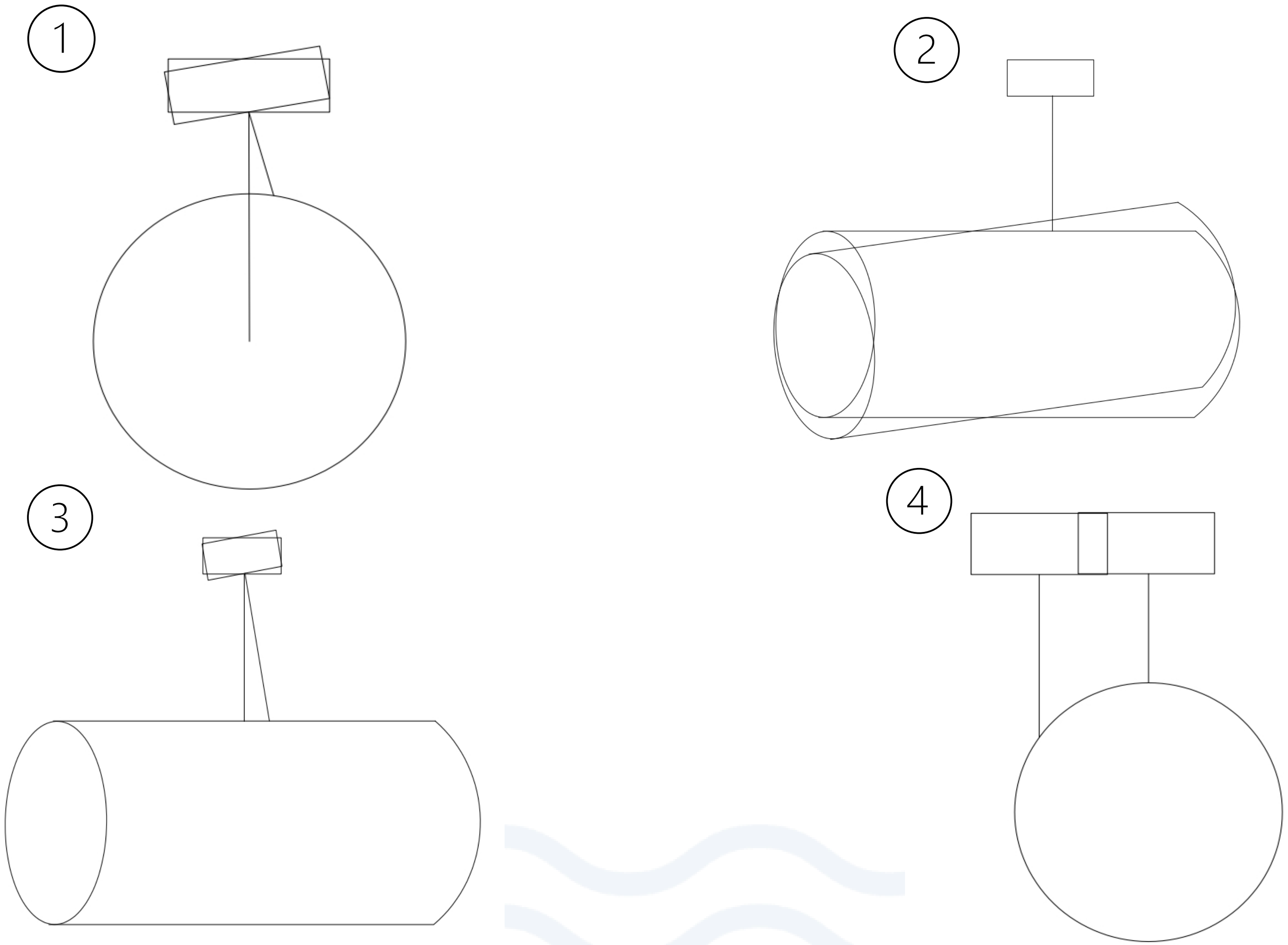
- Keyence IL-100
  - Firstly choose this model of wall thickness measurement machine.
  - After doing some research, this model could not satisfied the environment and condition in real application.
  - The measurement distance is a little bit short, the measurement distance is only 55mm, the machine may damaged by protruding pieces of iron
  - So after carefully consideration, then choose Keyence IL-300.

- Keyence IL-300
  - Keyence IL-300 ‘s measurement distance perfectly satisfied the real application, its measurement distance is 290mm.
  - As doing research deeply, this kind of model’s accuracy can not satisfy
  - Through communication with the supplier’s technical staff, it was learned that the accuracy of the IL-300 can only reach an estimated value of 0.1mm and cannot achieve the required accuracy.

- Keyence LK-400
  - Keyence LK-400’s accuracy can accurate achieve 0.1mm, and its measurement distance is 400±100mm, It can perfectly satisfied the real condition.

PAM SINCE 1856 穆松桥

Simulation of four scenarios that affect accuracy

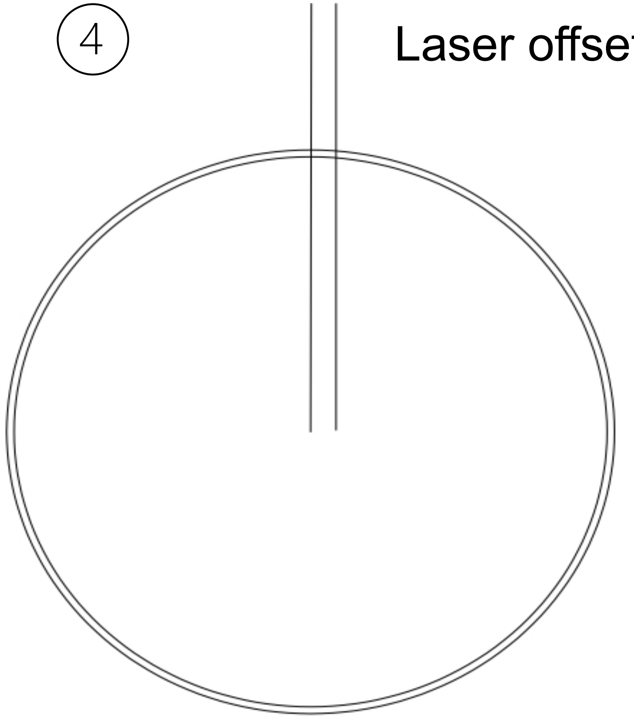


In the next slide, we simulated both cases of Fig. 1 and Fig. 4. Fig. 2 and Fig. 3 are simulated in essentially the same way as Fig. 1.(They are both angle shift)

MEASUREMENT METHODS FOR LARGE-DIAMETER PIPES AND EXPERIMENTS AFFECTING ACCURACY

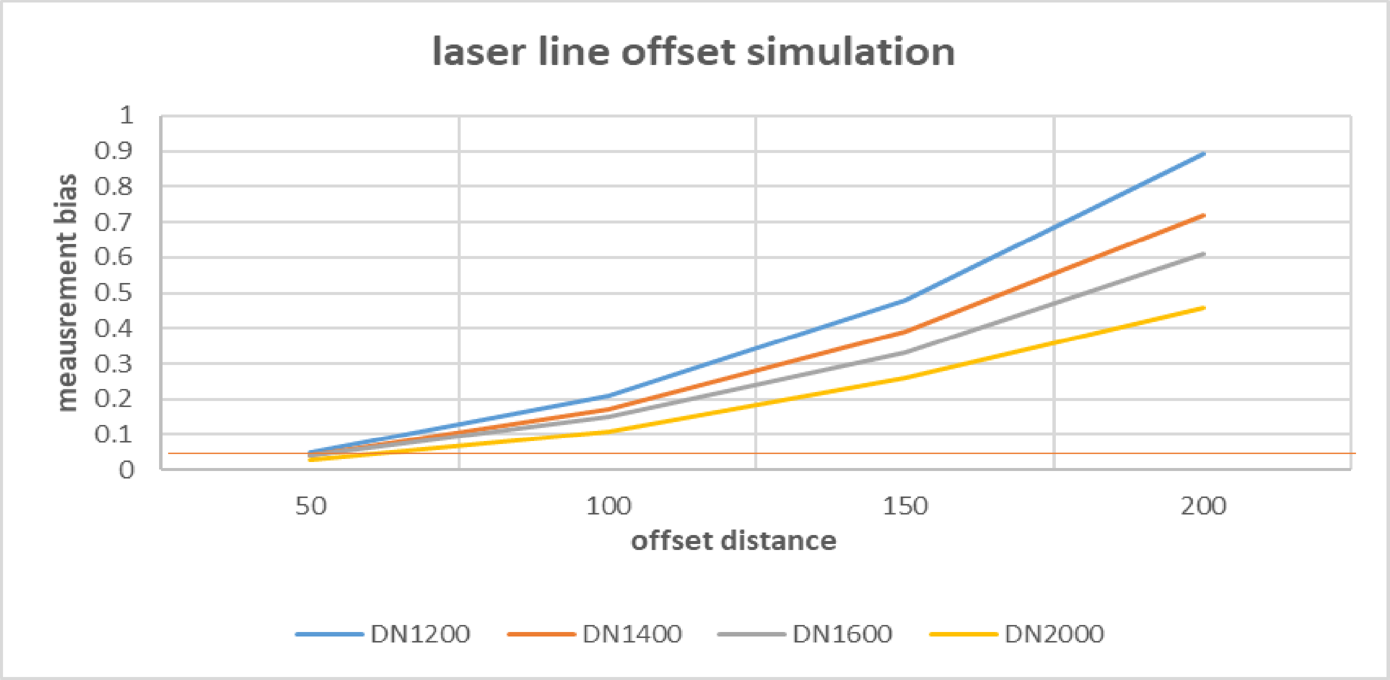
Simulation of four scenarios that affect accuracy

④ Laser offset



Formula:
Measured wall thickness after offset minus original wall thickness

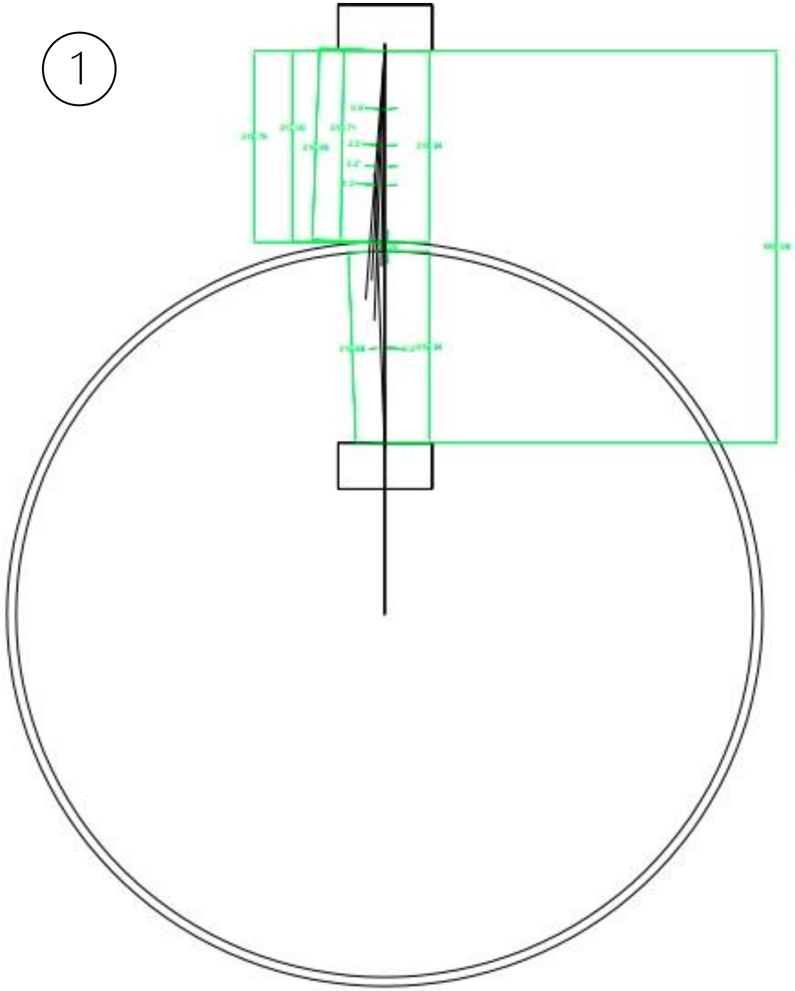
offset distance mm	DN1200	DN1400	DN1600	DN2000
50	0.05	0.04	0.04	0.03
100	0.21	0.17	0.15	0.11
150	0.48	0.39	0.33	0.26
200	0.89	0.72	0.61	0.46



When the offset distance is within 50mm, the resulting accuracy error is acceptable, all less than 0.1 mm.

Simulation of four scenarios that affect accuracy

①



For pipes DN1200, DN1400,DN1600, DN2000, when the laser deflection angle does not exceed 1.5 degrees, the error caused by the angular shift is within the acceptable range, all are less than 0.1mm.

→inclinometers are effective in detecting angular shifts

DN1200

angle of nevigation	offset distance
0.9	0.17
2.2	0.25
3.2	0.34
4.3	0.46

DN1400

angle of nevigation	offset distance
0.5	0.01
1	0.02
1.5	0.04
2	0.08

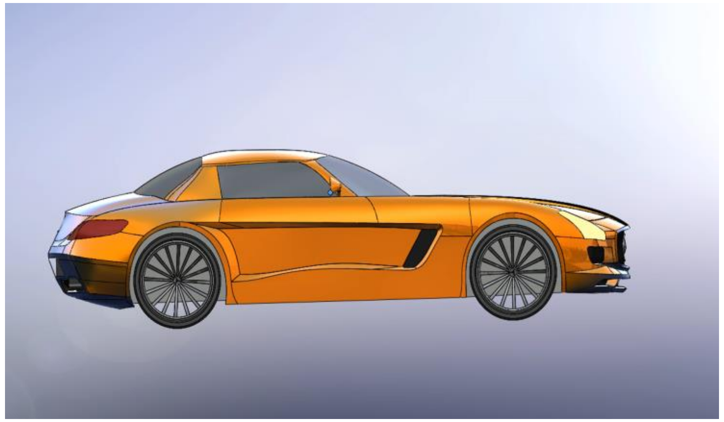
DN1600

angle of nevigation	offset distance
0.5	0
1	0.04
1.5	0.07
2	0.15

DN2000

angle of nevigation	offset distance
0.5	0.02
1	0.03
1.5	0.06
2	0.19

Simulation of four scenarios that affect accuracy



Tilt

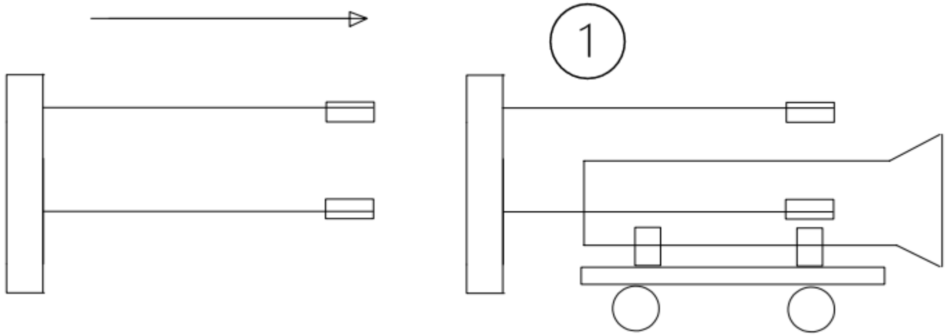
Roll angle

Azimuth

→ Inclinometers use the North Micro brand

MEASUREMENT METHODS FOR LARGE-DIAMETER PIPES AND EXPERIMENTS AFFECTING ACCURACY

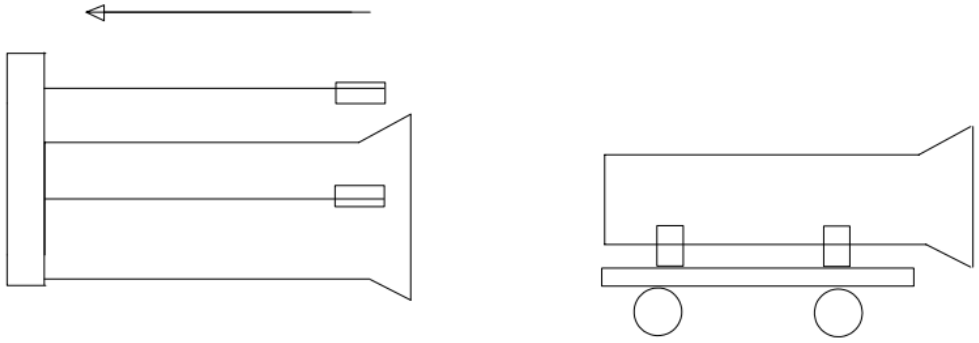
Three ways to measure large DN pipes



1

- In picture 1, the pipe is stationary, the sensors move from left to right.

- The laser sensor moves at a constant speed from side to side and the pipe is stationary after entering the measured area.



2

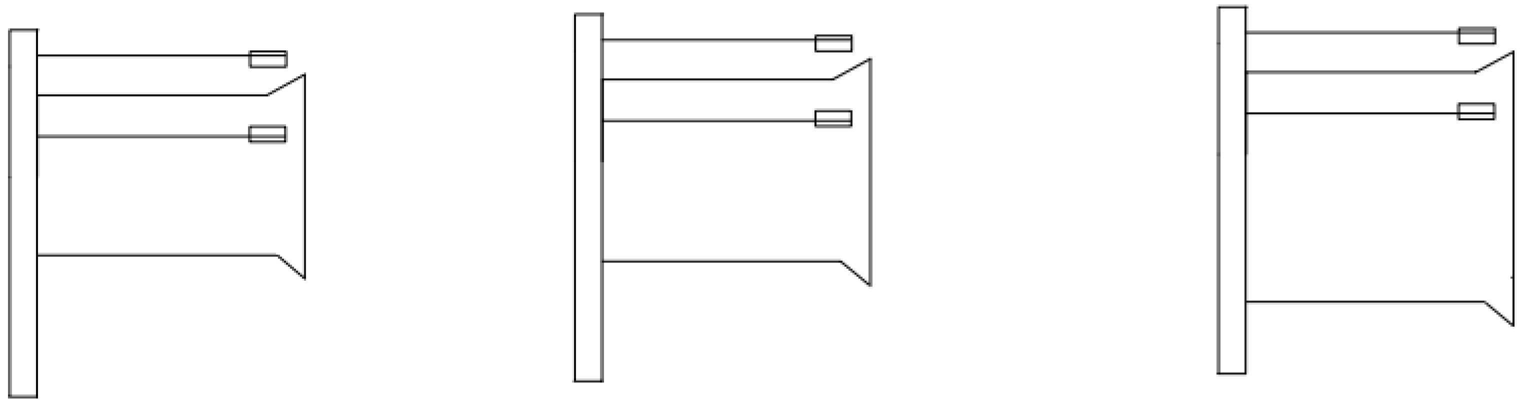
- In picture2, the sensors are stationary, the pipe move from right to left.

Three ways to measure large DN pipes

3

The pipe is stationary and the laser thickness measurement machine on both sides are moving as a whole at a constant speed.

Measurement of different pipe diameters



For the measurement of pipes of different diameters, the upper wall of pipe is fixed in position and a lifting device is installed on the trolley so that the pipe can be freely adjusted up and down.

MEASUREMENT METHODS FOR LARGE-DIAMETER PIPES AND EXPERIMENTS AFFECTING ACCURACY

SINCE 1856
穆松桥

Summary

- Of the three methods mentioned above, I think the best method is method 3.

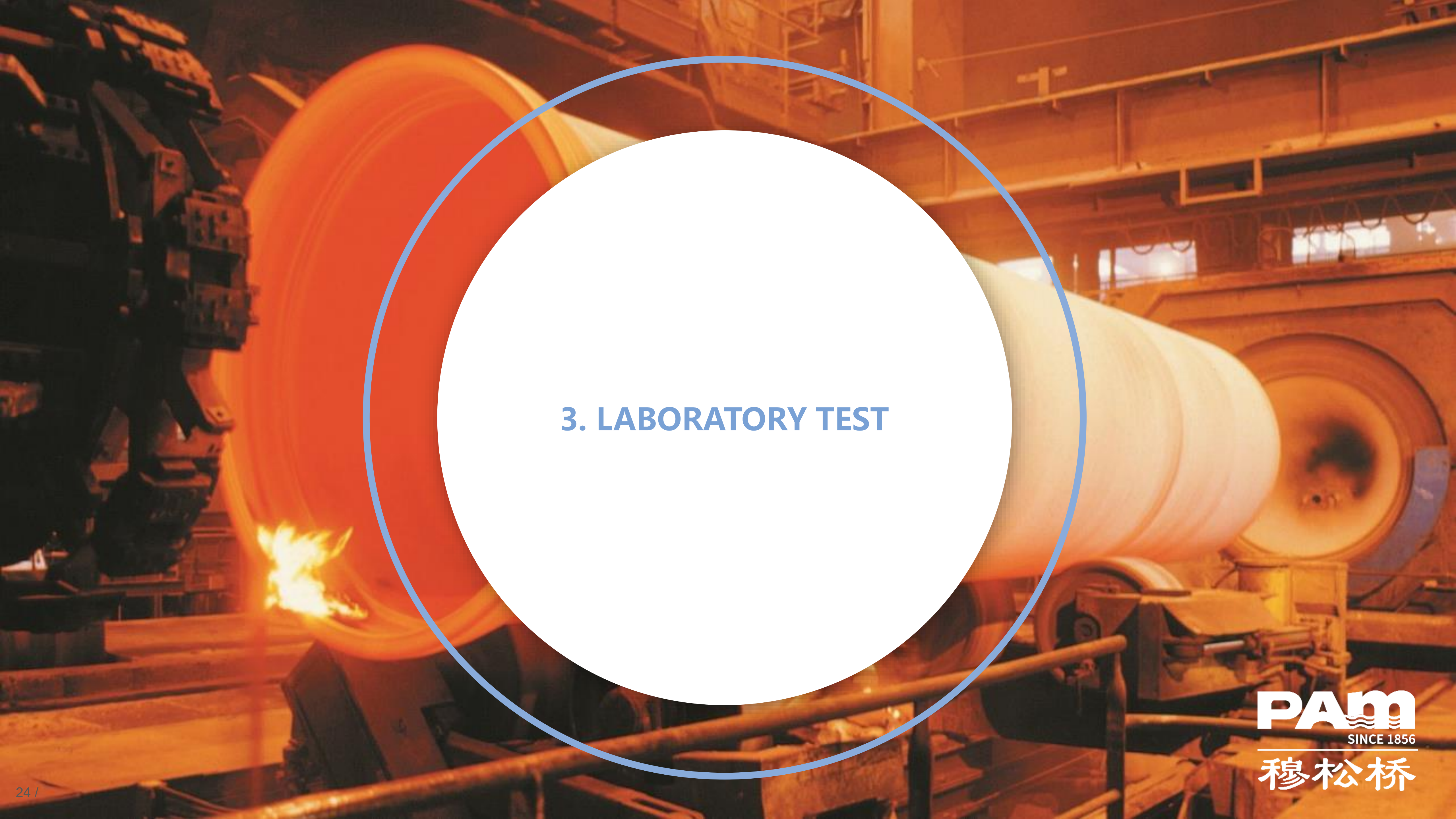
- The three meter length on each side allows for greater stability of the laser sensor. But in method 3 the cost may be much higher than method 1 and 2, because in method 3 need 4 laser measurement machine.

- We need to physically inspect the factory site to see if the entire machine can be installed.

- Method 1 and 2 require beams up to more than six meters in length, and when there is some vibration at one end of the beam, the laser measurement machine will vibrate more strongly at that end, increasing the measurement deviation.

- Need to know how much time of the up and down stream line. Also we should know how much time that measure one pipe needed.

3. LABORATORY TEST



LABORATORY TEST (110KCNY INVESTED)

Pipe under test

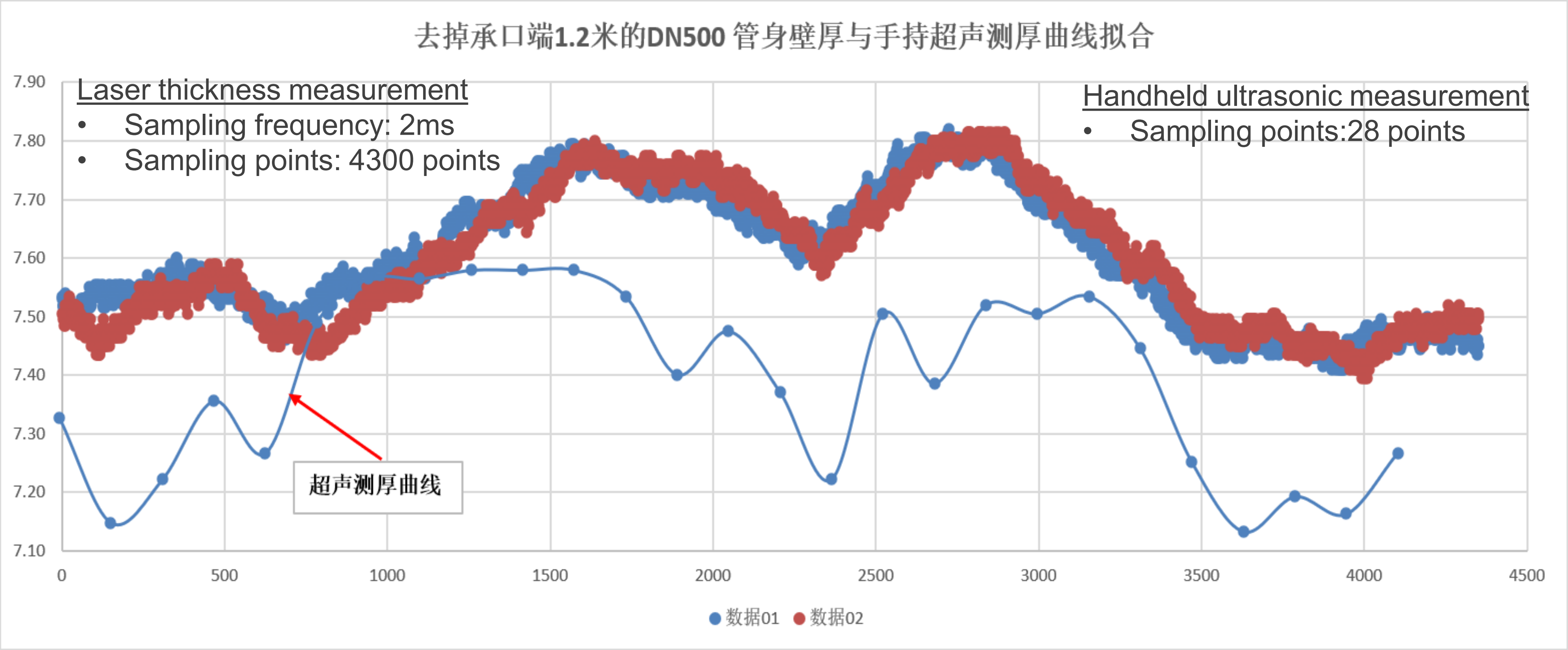
LABORATORY TEST (110KCNY INVESTED)

Pre-test plan

Test date	22/11/30 afternoon	
Test site	Maanshan pipe casting machine	
Measurment method	Laser thickness measurement	handheld ultrasonic thickness measurement
Pipe type	DN500 K9	
Pipe direction	Entering the measurement from the socket	
Measuring distance	3.8m	
Test number	2	1
Sampling frequency	2ms	/
Sampling points	4300	28

LABORATORY TEST (110KCNY INVESTED)

Pre-test results



- The 2 curves formed by the laser thickness measurement data are basically the same, which means that repeatability is quite good
- The difference between the curve formed by laser and handled ultrasonic is around max 0.4mm, which means that accuracy seems to be good, but to be confirmed by more tests

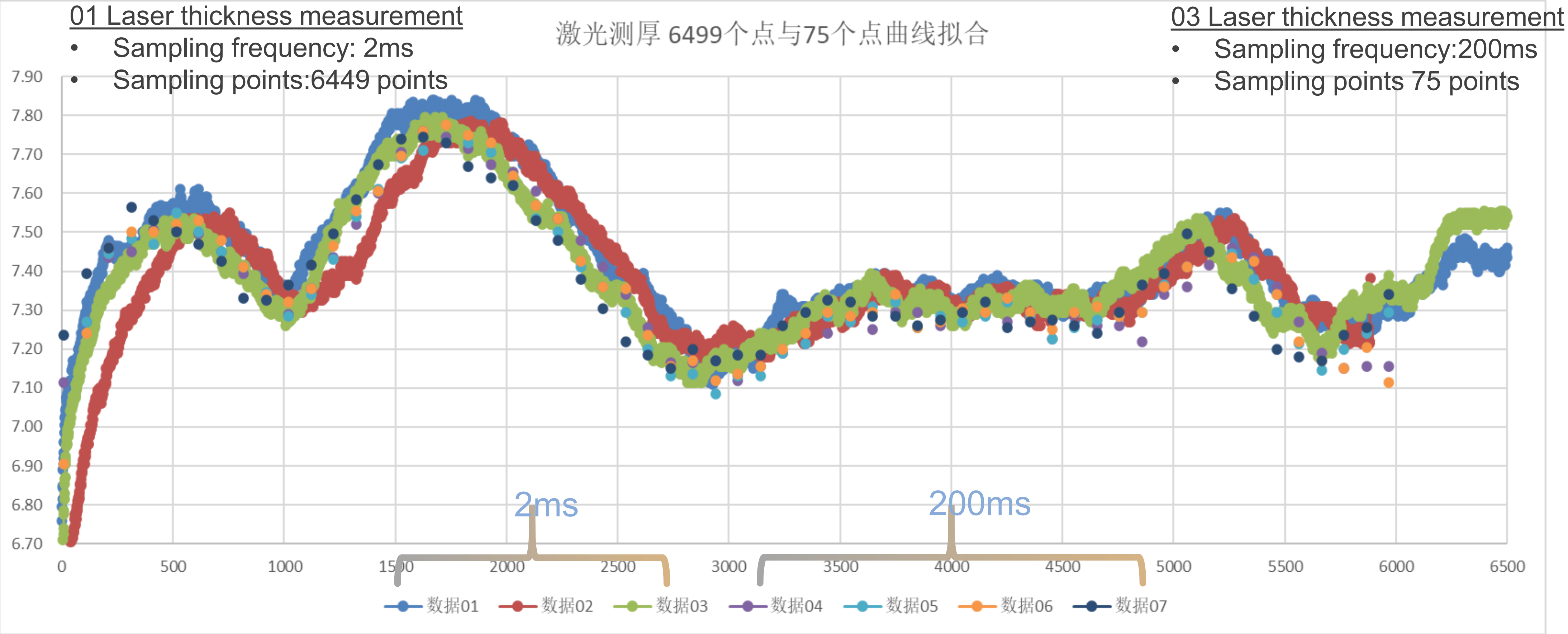
LABORATORY TEST (110KCNY INVESTED)

Test plan

Test date	22/12/01 afternoon	22/12/01 afternoon	22/12/02 afternoon	22/12/01 afternoon
Test site	Maanshan pipe casting machine			
Measurment method	01 Laser thickness measurement	02 Laser thickness measurement	03 Laser thickness measurement	handheld ultrasonic thickness measurement
Pipe type	DN500 K9			
Pipe direction	Entering the measurement from the spigot			
Measuring distance	3.8m			
Test number	3	3	4	1
Sampling frequency	2ms	20ms	200ms	/
Sampling points	6499	649	75	37

LABORATORY TEST (110KCNY INVESTED)

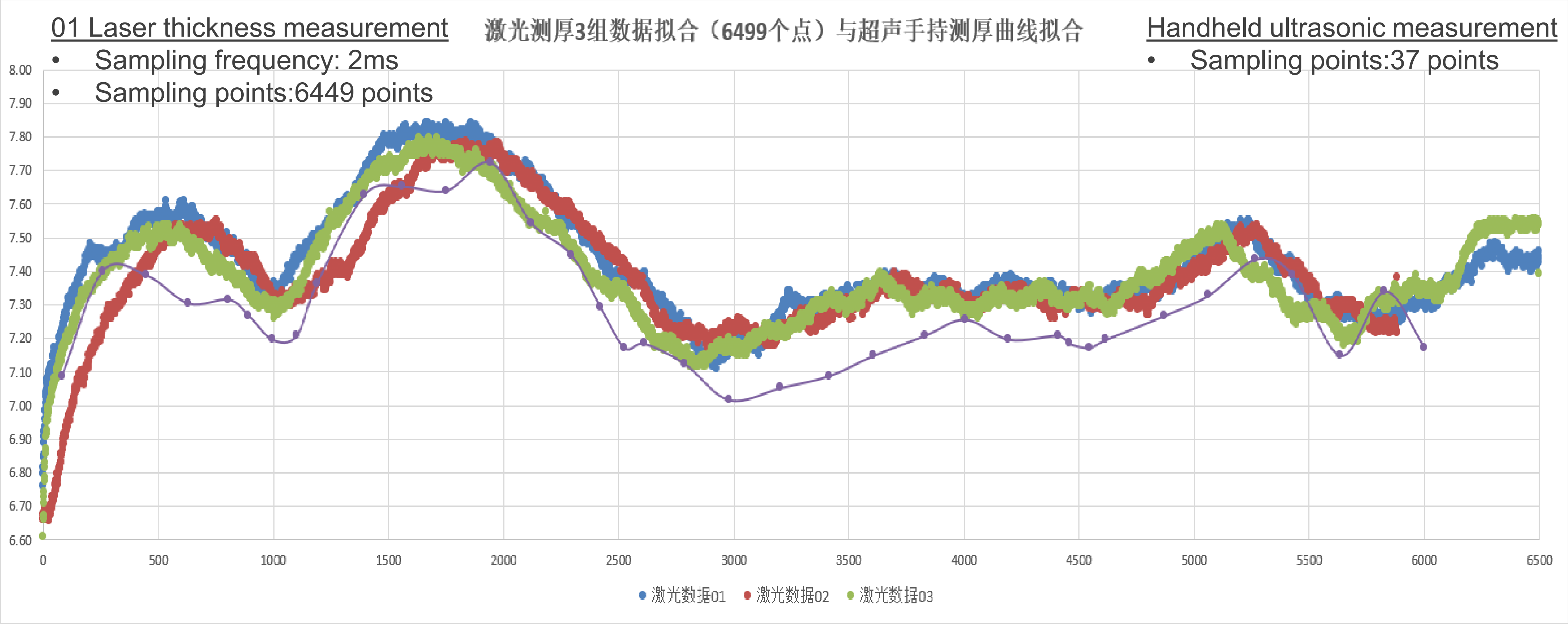
Test results



- It can be seen that the curves formed by 2ms and 200ms are basically the same, indicating that sampling frequency difference has no effect on the accuracy

LABORATORY TEST (110KCNY INVESTED)

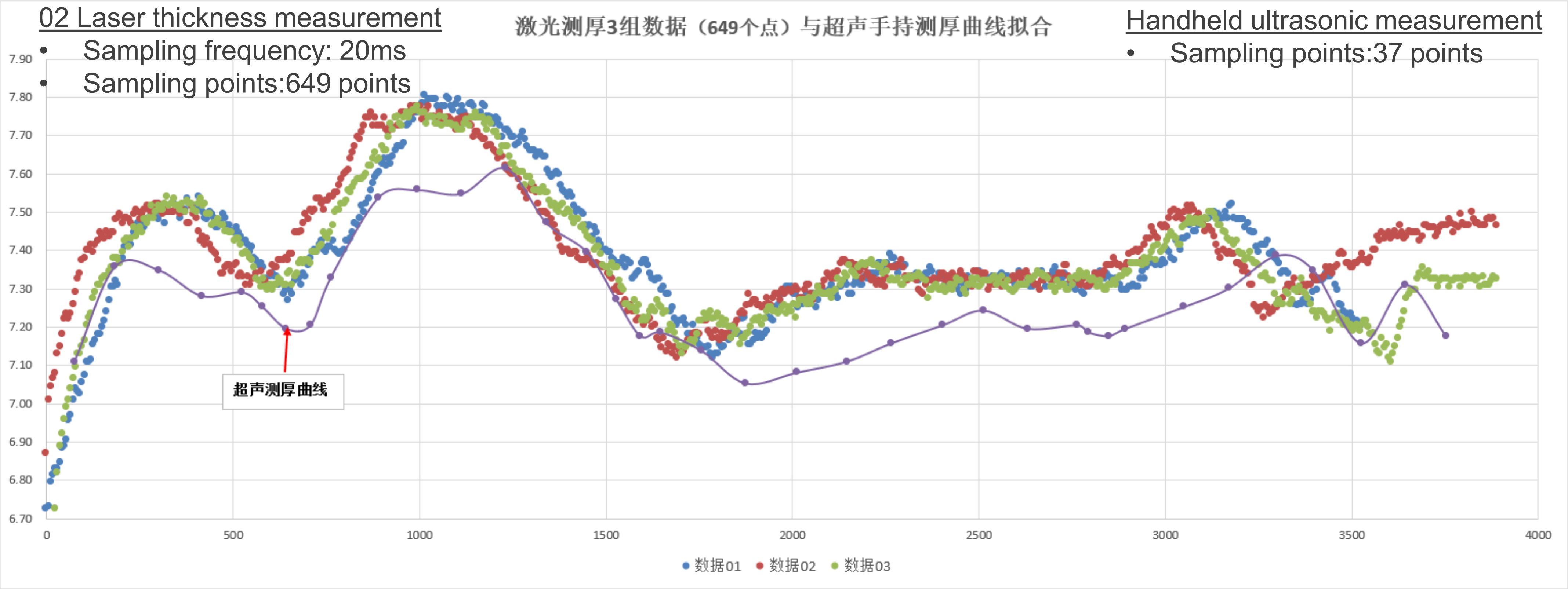
Test results



LABORATORY TEST (110KCNY INVESTED)

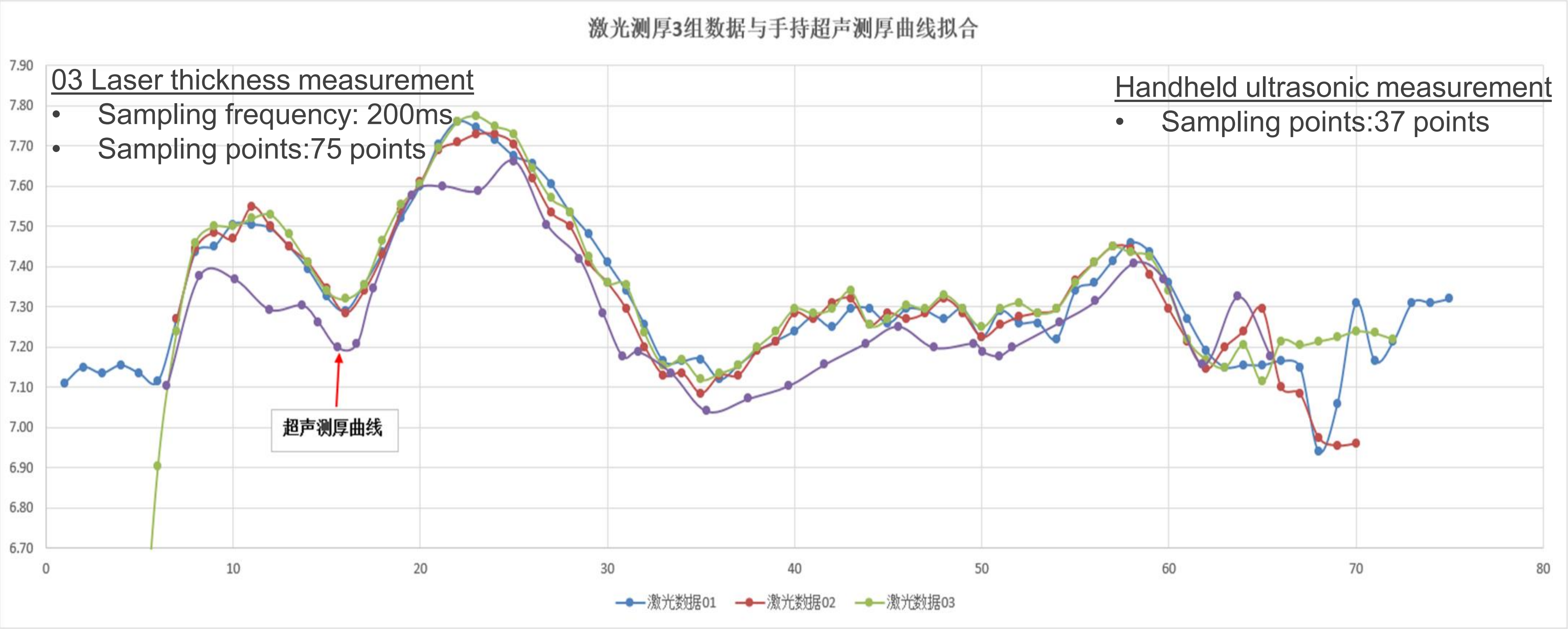
SINCE 1856
穆松桥

Test results



LABORATORY TEST(110KCNY INVESTED)

Test results



- From the above 3 figures from slide 30 to 32, it can be concluded that the curves formed by the 2ms,20ms, 200ms are basically the same as the curve formed by handheld ultrasonic thickness measurement, with a maximum deviation of about 0.2mm
- It should be finally calibrated by micrometer, which is the most accurate measurement method

LABORATORY TEST(110KCNY INVESTED)

Test plan

Test date	22/12/01 afternoon	22/12/01 afternoon	22/12/02 afternoon	22/12/01 afternoon	/
Test site	Maanshan pipe casting machine				
Measurment method	01Laser thickness measurement	02 Laser thickness measurement	03 Laser thickness measurement	handheld ultrasonic thickness measurement	micrometer
Pipe type	DN500 K9				
Pipe direction	Entering the measurement from the spigot				
Measuring distance	3.8m				
Test number	3	3	3	1	1
Sampling frequency	2ms	20ms	200ms	/	/
Sampling points	6499	649	75	37	38

LABORATORY TEST(110KCNY INVESTED)

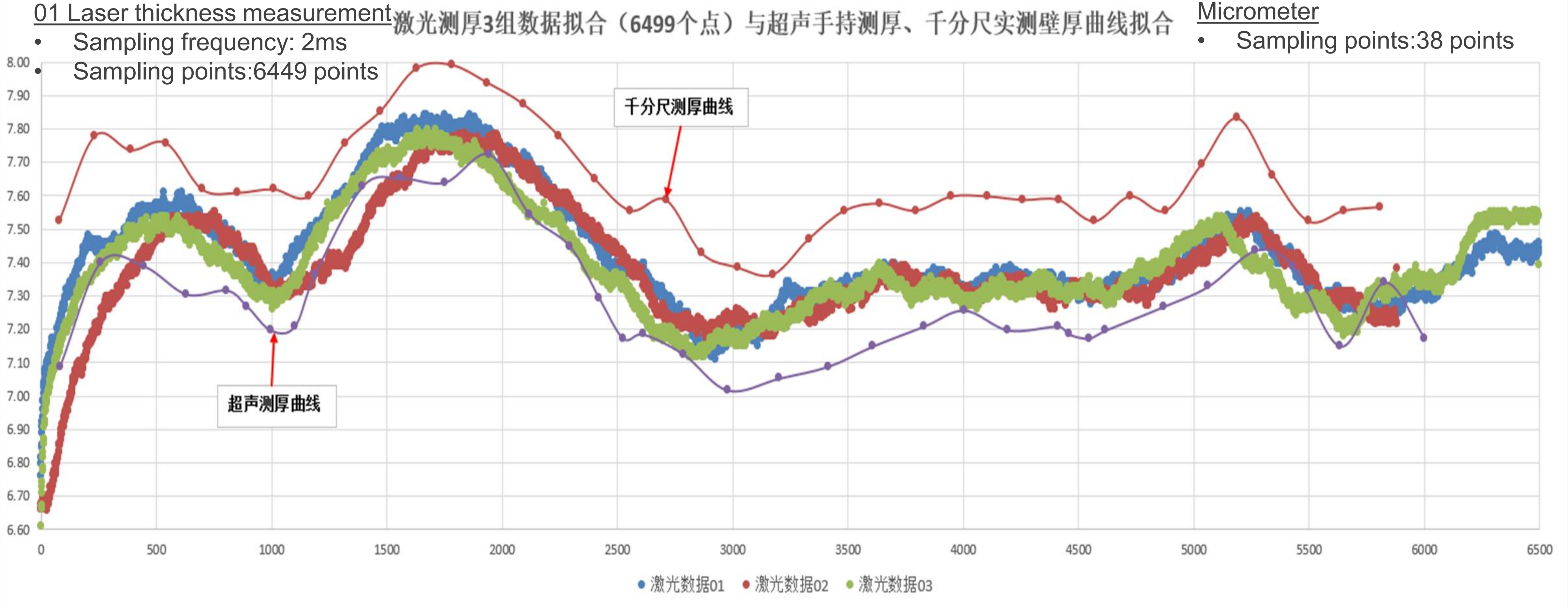
Test results

01 Laser thickness measurement

- Sampling frequency: 2ms
- Sampling points:6449 points

Micrometer

- Sampling points:38 points



LABORATORY TEST(110KCNY INVESTED)

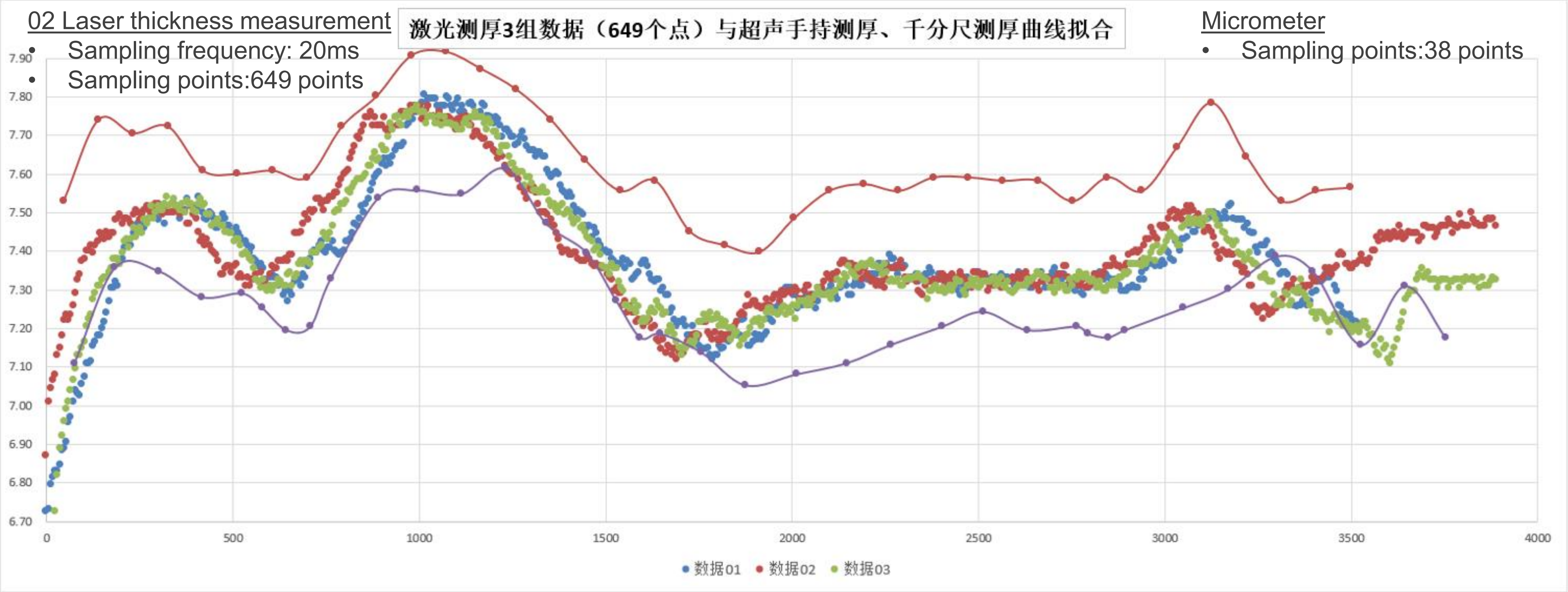
Test results

02 Laser thickness measurement

- Sampling frequency: 20ms
- Sampling points:649 points

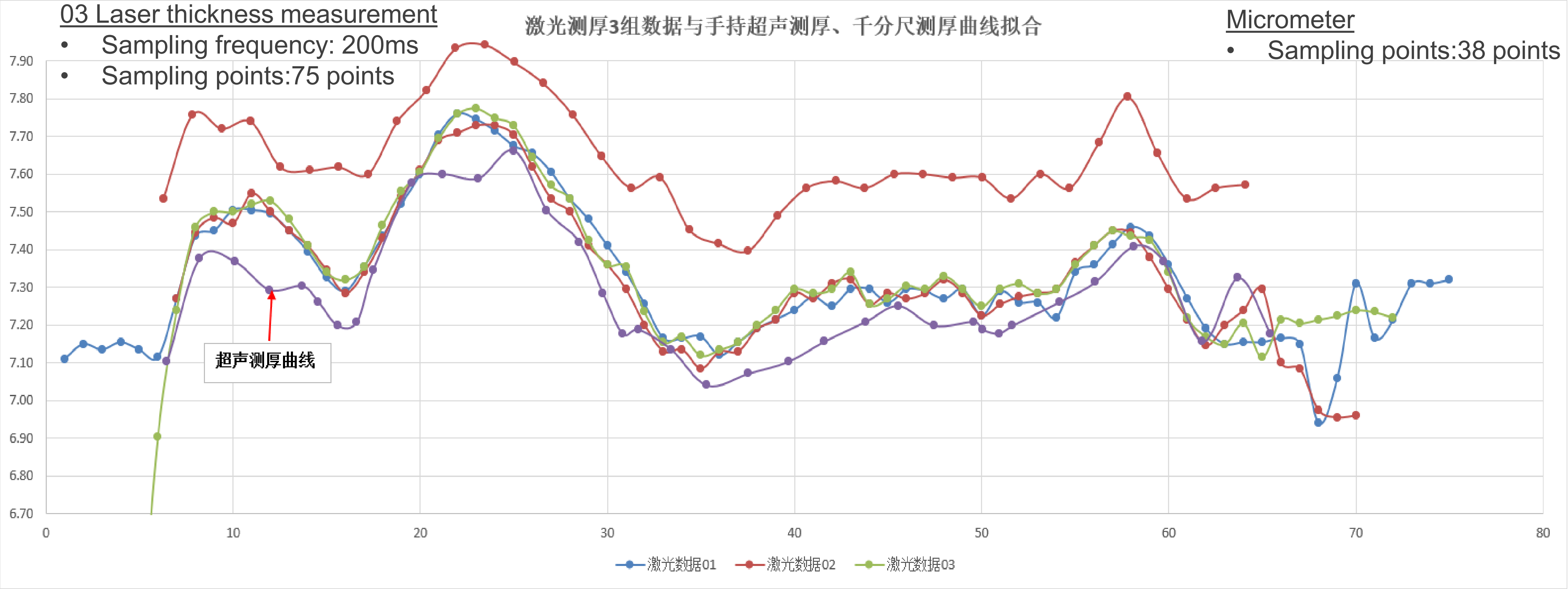
Micrometer

- Sampling points:38 points



LABORATORY TEST(110KCNY INVESTED)

Test results



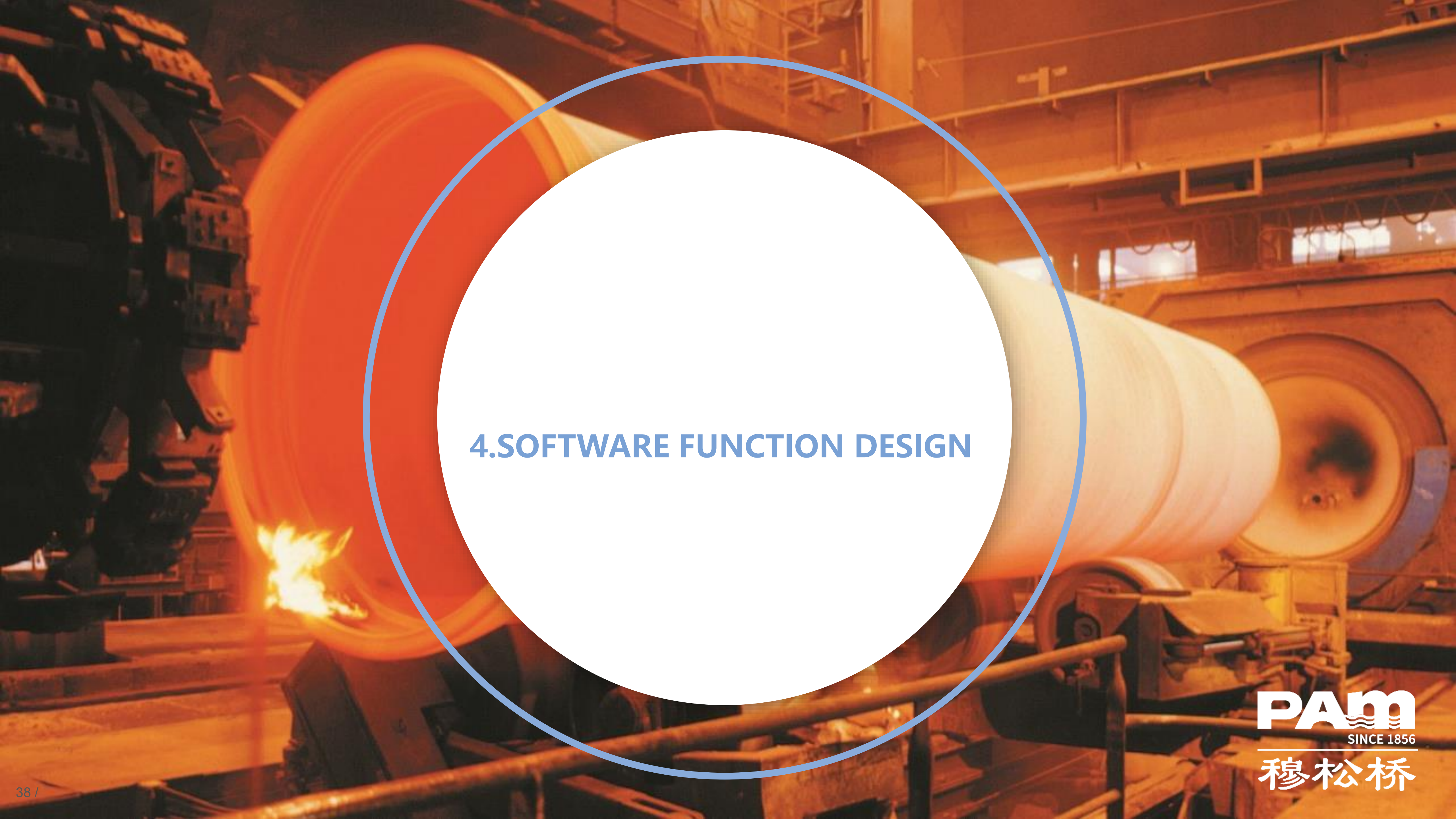
From the above 3 figures from slide 34 to 36, it can be concluded that the curve formed by 2ms, 20ms, 200ms is basically the same as the curve formed by the 38 points measured by micrometer, with a maximum deviation of about 0.25mm.

LABORATORY TEST(110KCNY INVESTED)

Conclusion

- From the above experiments, it can be found that the wall thickness curves measured by micrometer, ultrasonic handheld thickness measurement and laser thickness measurement are basically the same

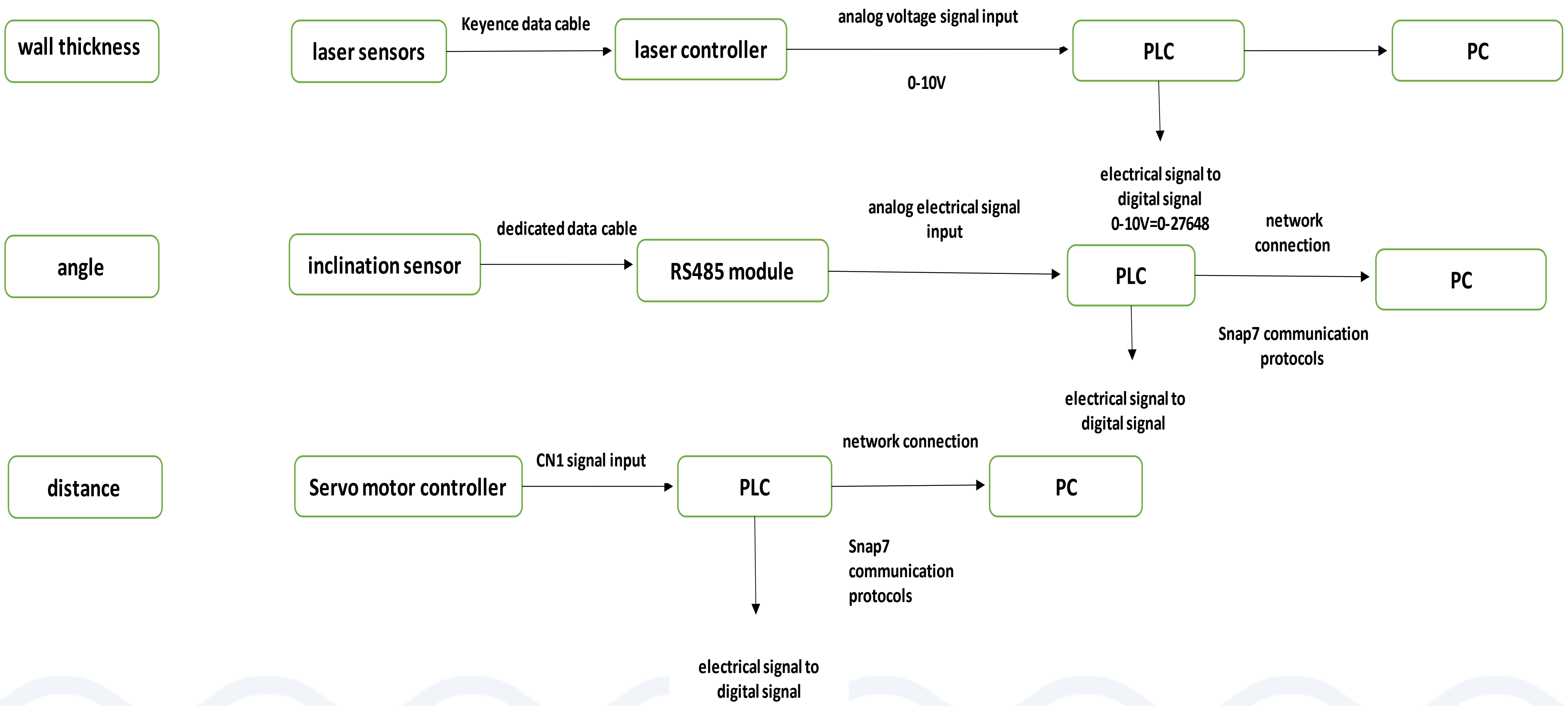
- The conclusion proves the feasibility of laser thickness gauge in large diameter pipe thickness measurement and lays the foundation for the subsequent production of the test machine.



4.SOFTWARE FUNCTION DESIGN

SOFTWARE FUNCTION DESIGN

Thickness measurement machine data



SOFTWARE FUNCTION DESIGN

Measurement main screen

DN

Thickness standard

Connector

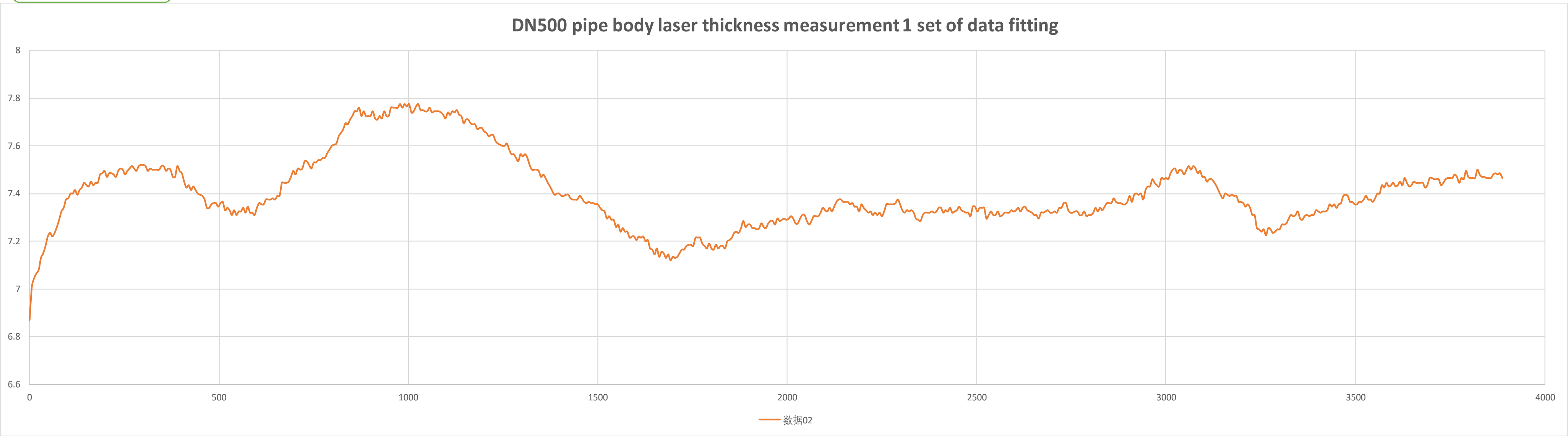
Group of team

Pipe number

Date

Piping grade

Wall thickness curve



Minimum wall thickness

Maximum wall thickness

Average wall thickness

Inclination sensor

Upper side:

Lower side:

X

Y

Z

Status

OK/NOK

OK/NOK

Measuring time

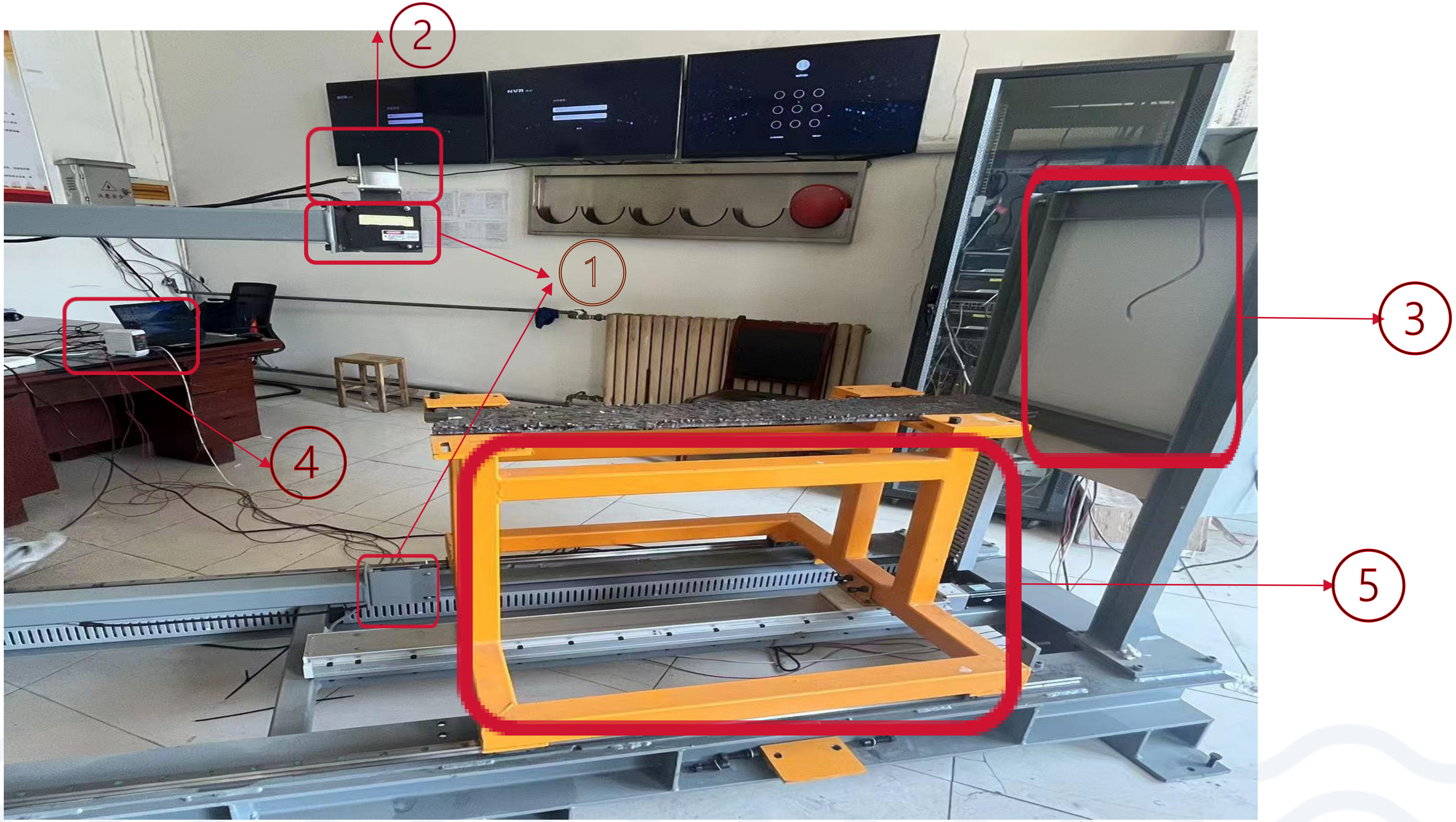
Measurement points

5.INTERMEDIATE INDUSTRY TEST

PAM SINCE 1856 穆松桥

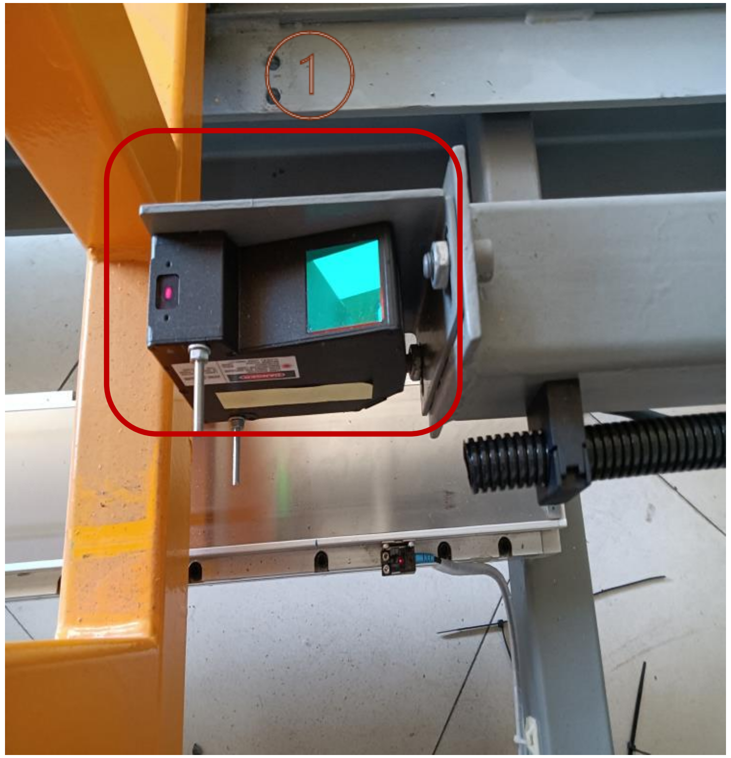
INTERMEDIATE INDUSTRY TEST

Introduction of tester



INTERMEDIATE INDUSTRY TEST

Introduction of tester



Upper and lower laser thickness measurement machine for pipe thickness measurement

INTERMEDIATE INDUSTRY TEST

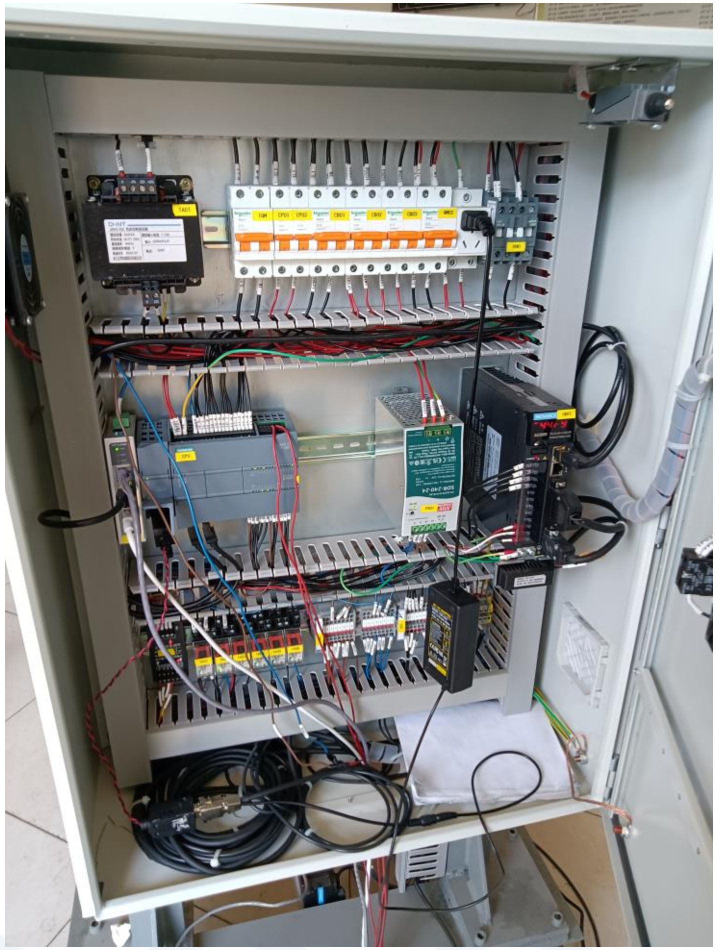
Introduction of tester

Inclinometer for detecting vibration amplitude of laser thickness measurement machine

INTERMEDIATE INDUSTRY TEST

Introduction of tester

3



The machine operator control box is used to control the operation of the laser thickness measurement machine and the trolley as well as the data processing.

INTERMEDIATE INDUSTRY TEST

Introduction of tester

- The display of the laser thickness measurement machine is used for real- time display of the thickness measurement data.

- The upper and lower numbers correspond to the distance of the upper and lower sensors from the upper and lower walls of the pipe.

Introduction of tester

5

Trolley —— The pipe is measured by moving the trolley back and forth.

INTRODUCTION OF THE TESTER AND THE SOFTWARE INTERFACE

PAM SINCE 1856 穆松桥

Tester operation video

INTERMEDIATE INDUSTRY TEST

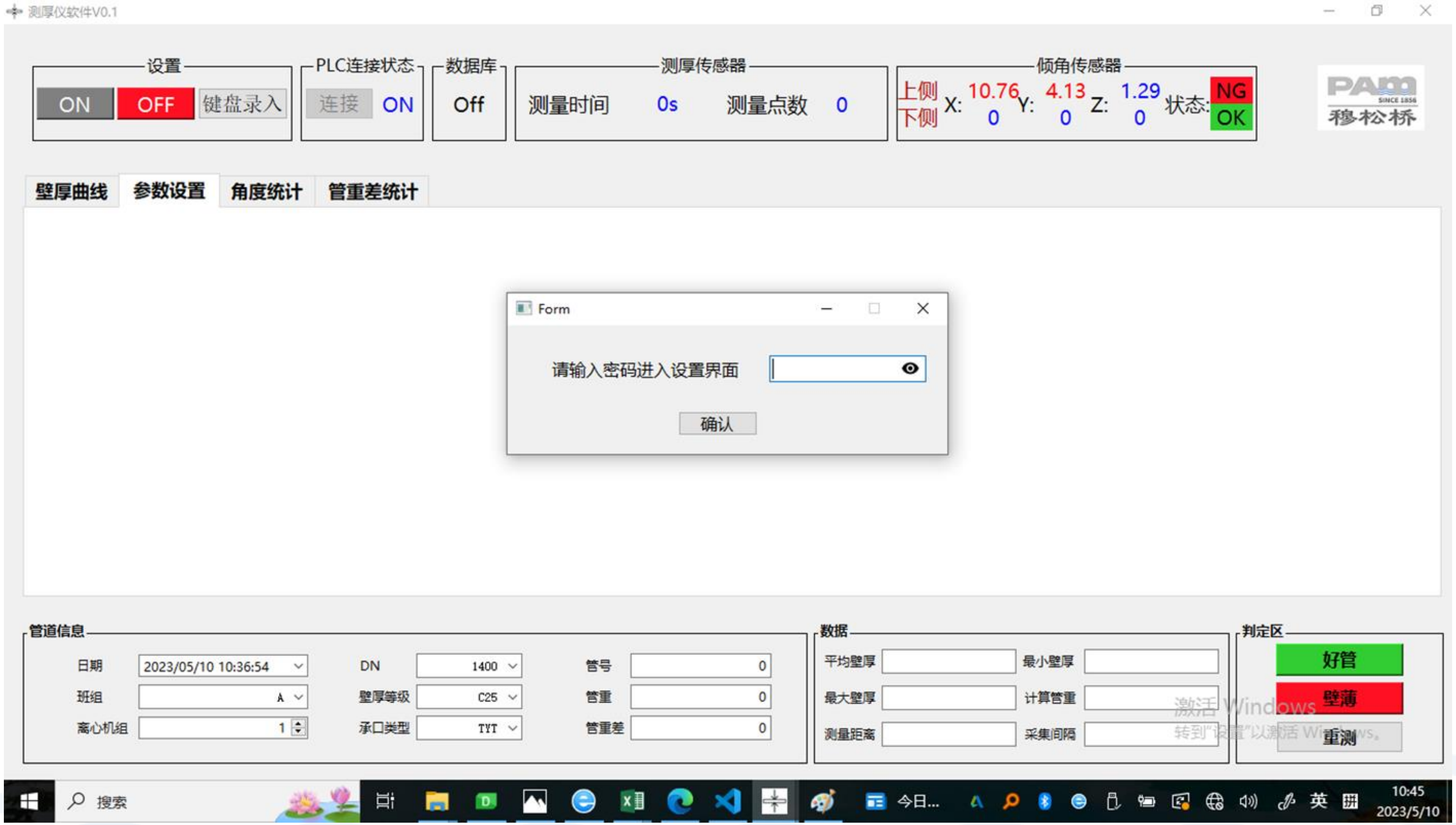
Introduction of interface

Measurement points: number of points collected during measurement

1.calculation of pipe weight: weight of socket +weight of pipe body(volume *density calculated from measured wall thickness)
2.Measurement distance: measured pipe distance
3. Acquisition interval: distance between 2 points.

INTERMEDIATE INDUSTRY TEST

Introduction of interface



Password：

PAM123!

Enter the above password and enter the parameter interface

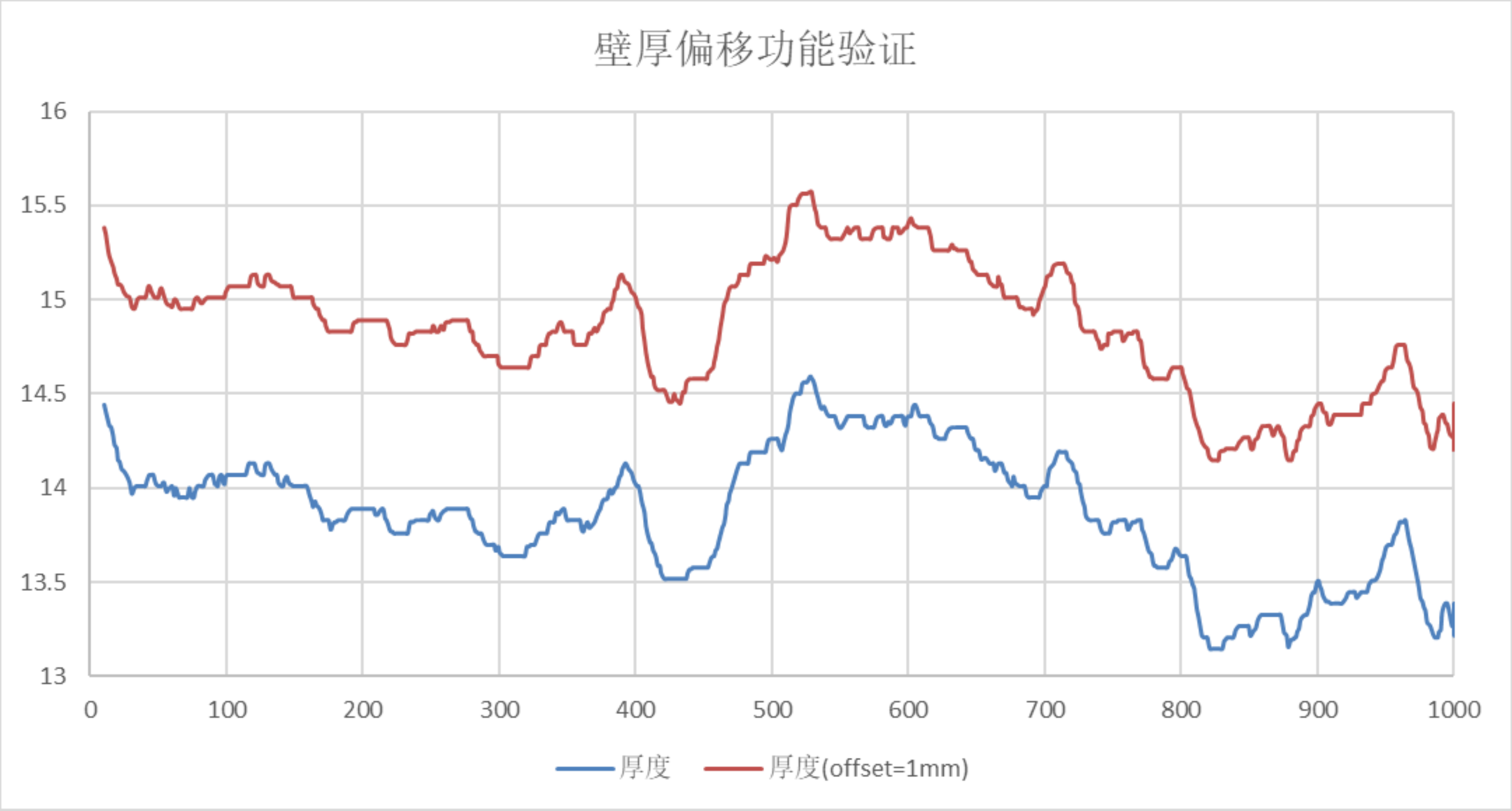
INTERMEDIATE INDUSTRY TEST

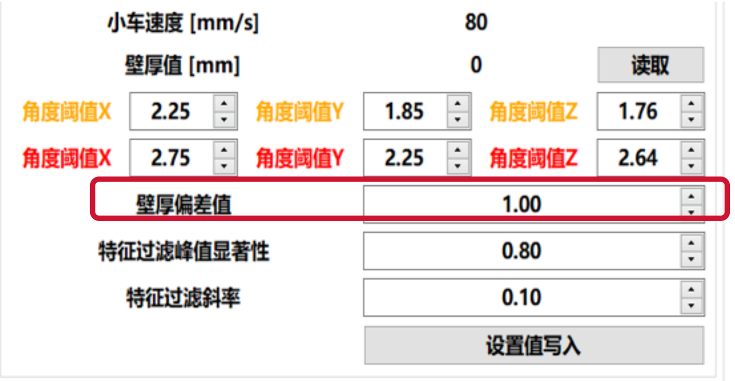
Introduction of interface

Wall thickness deviation value: the wall thickness curve can be adjusted as a whole (up or down) to make it close to the actual wall thickness curve.

INTERMEDIATE INDUSTRY TEST

Introduction of interface





After setting a wall thickness deviation value of 1mm, the left wall thickness curve is adjusted upwards by 1mm as a whole.

INTERMEDIATE INDUSTRY TEST

Introduction of interface

The red box is the area to fill in the angle thresholds

- Yellow is the threshold for the yellow area
- Red is the threshold for the red area

Impact is controlled at 0.1mm

- Angle X threshold– yellow value is set at 1.35°
- Angle Y threshold- yellow value is set at 1.1°
- Angle Z threshold- yellow value is set at 0.6°

INTERMEDIATE INDUSTRY TEST

Introduction of interface

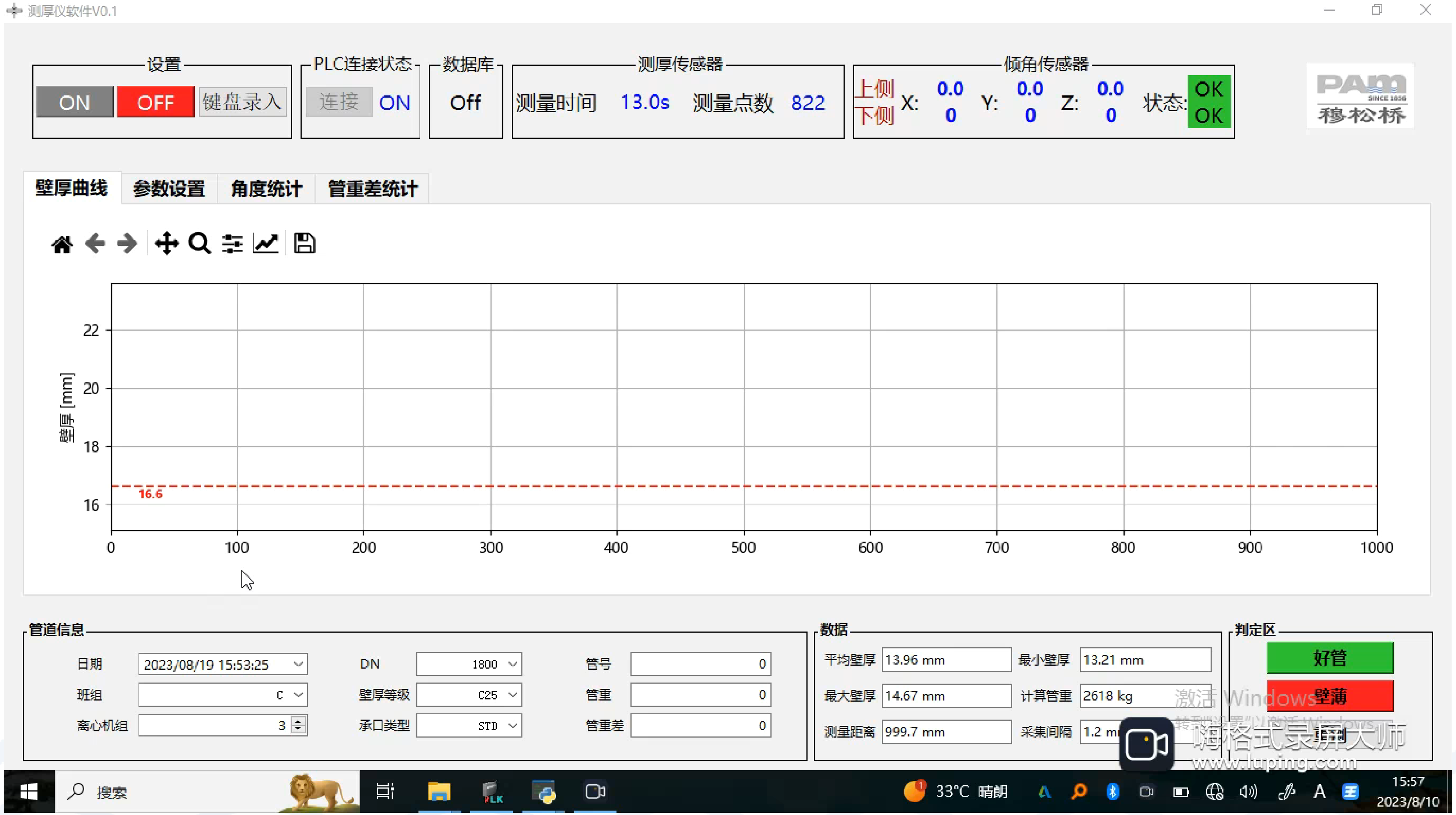
- Feature filter peak significance: Indicates the sudden height distance of the peak, such as the picture means the sudden height distance of the peak 0.8mm
- Feature filter slope: Indicates the degree of slope of the peak, the steeper the value is the larger.

INTERMEDIATE INDUSTRY TEST

Demonstration of page operation

INTERMEDIATE INDUSTRY TEST

Demonstration of page operation



INTERMEDIATE INDUSTRY TEST

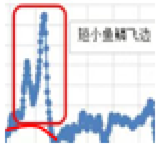
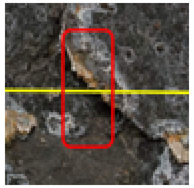
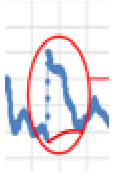
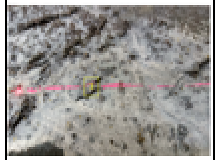
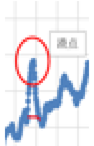
Demonstration of page operation

INTERMEDIATE INDUSTRY TEST

Samples test and statistic summary of inner surface

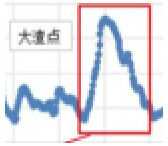
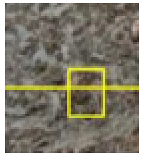
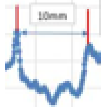
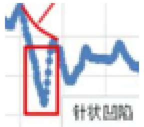
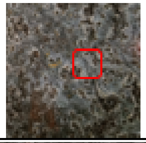
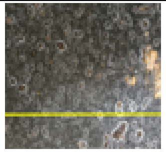
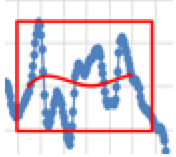
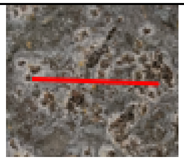
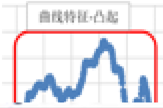
PAM SINCE 1856 穆松桥

热模内壁壁厚曲线特征

序号	内壁特征	曲线特征	照片	拟合处理方案	处理依据	特征尺寸（长*宽*高）	数据处理的依据	承口-特征比例（范围200*200mm）	管身-特征比例（范围200*200mm）	插口-特征比例（范围200*200mm）	备注
1	短小的鱼鳞纹飞边							5-10%	0-5%	70-80%	
2	宽大的鱼鳞纹飞边			从根部拟合符合正常壁厚趋势的曲线，去掉波峰（如图红色拟合线）	1、飞边不算实际壁厚； 2、飞边的底部为实际壁厚，因此去掉典型的突高曲线，取根部拟合。	曲线数据：突高高度>1mm，突高持续<1mm，突高向下持续宽度>2mm；定义为飞边，拟合方式采用底部曲线平滑拟合	定义飞边的宽度、高度，超过定义为飞边，采用底部曲线拟合	0-5%	0-5%	5-10%	
3	长翘的鱼鳞纹飞边							50-60%	0-5%	20-30%	
4	小渣点			曲线中为渣点特征，拟合壁厚曲线时，建议去掉波峰，从根部拟合连线，（如图红色拟合线）	小渣点高度约0.5毫米，钉扎在管道内壁并不牢固，极易磕掉	曲线数据：突高>0.5mm，突高持续<1mm，宽度≤2mm；定义为小渣点，拟合方式采用底部曲线平滑拟合	定义飞边的宽度、高度，超过定义为飞边，采用底部曲线拟合	20~30%	30~40%	80~90%	

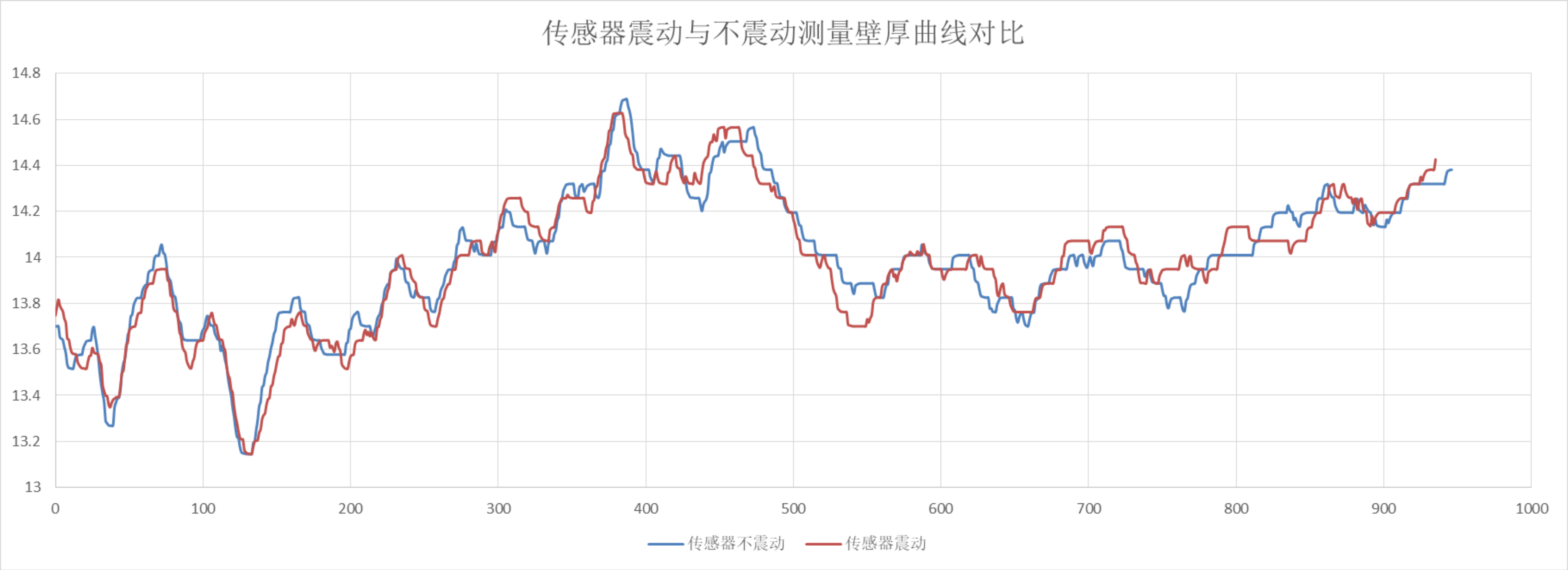
INTERMEDIATE INDUSTRY TEST

Samples test and statistic summary of inner surface

5	大渣点			曲线中为大渣点特征，建议保留曲线	这种渣点很难磕掉，结合在内壁很牢固，卡尺测量渣层也会计算到厚度	曲线数据：突高>0.5mm，宽度（直径）>2mm；定义为大渣点，拟合方式为保留曲线平滑拟合	定义渣点的直径、高度，超过定义为飞边，采用底部曲线拟合	80–90%	10~20%	80~90%		wan
6	宽大凹陷			凹陷特征为实际壁厚特征，因此建议保留实际壁厚特征曲线。	因为使用千分尺和游标卡尺实际测量可以测到底部实际值	曲线数据：凹陷>0.5mm，宽度≥3mm；定义为大凹陷，拟合方式采用底部曲线平滑拟合	定义凹陷的宽度、深度，超过定义为宽的凹陷，保留曲线	5-10%	5-10%	5-10%		
7	针状凹陷			图中针孔凹陷，拟合曲线时，建议去掉，从上根部拟合	因为实际卡尺测量无法测量到底部。	曲线数据：凹陷>0.5mm，宽度<3mm；定义为针状凹陷，拟合方式采用上部曲线平滑拟合	定义凹陷的宽度、深度，超小于的定义为针状的凹陷，拟合曲线	20-30%	5-10%	20-30%		
8	小渣层			小渣层曲线特征，建议保留原始曲线	因为使用千分尺和游标卡尺实际测量可以测到渣的顶部与底部的实际值	曲线数据：曲线波动（高低范围）在0~0.5mm，曲线平滑拟合	定义渣层的宽度、深度，如何保留曲线	20-30%	80%	100%		
9	大渣层			大渣层曲线特征，建议拟合曲线按照大小渣点规律拟合	这种渣层很难磕掉，结合在内壁很牢固，卡尺测量渣层也会计算到厚度	按照大小渣点过透规律拟合曲线	按照大小渣点过透规律拟合曲线	70-80%	5-10%	50-60%		
10	凸起			凸起曲线特征，建议保留原始曲线	凸起是铁的实际隆起，因此保留实际曲线，卡尺测量也会测到实际的隆起厚度	曲线数据：突高≥1mm，突高持续（从低点到顶点）≥2mm，隆起的宽度≥3mm	定义隆起的宽度、高度，什么特征确定为隆起	5-10%	5-10%	5-10%		
10	水流纹			水流纹曲线特征，建议保留原始曲线	水流纹凸起是铁的实际隆起，因此保留实际曲线，卡尺测量也会测到实际的隆起厚度	截图(Alt + A)也有突高，[illegible]两侧不平齐，可以按照飞边处理	定义隆起的宽度、高度，什么特征确定为隆起	20-30%	100%	5-10%		

INTERMEDIATE INDUSTRY TEST

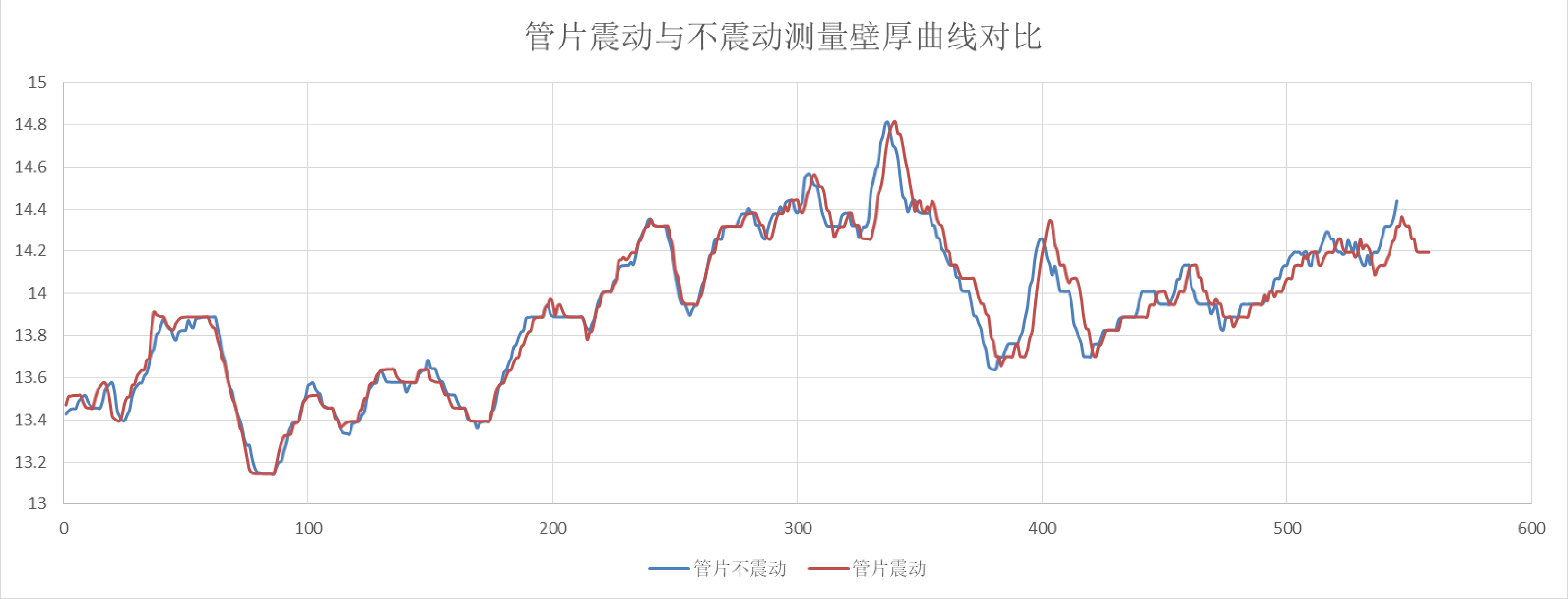
Vibration test



The angle change in vibration is recorded by the inclinometer, and the effect on the curve is shown above, the maximum influence is about 550mm, the impact is about 0.5mm

INTERMEDIATE INDUSTRY TEST

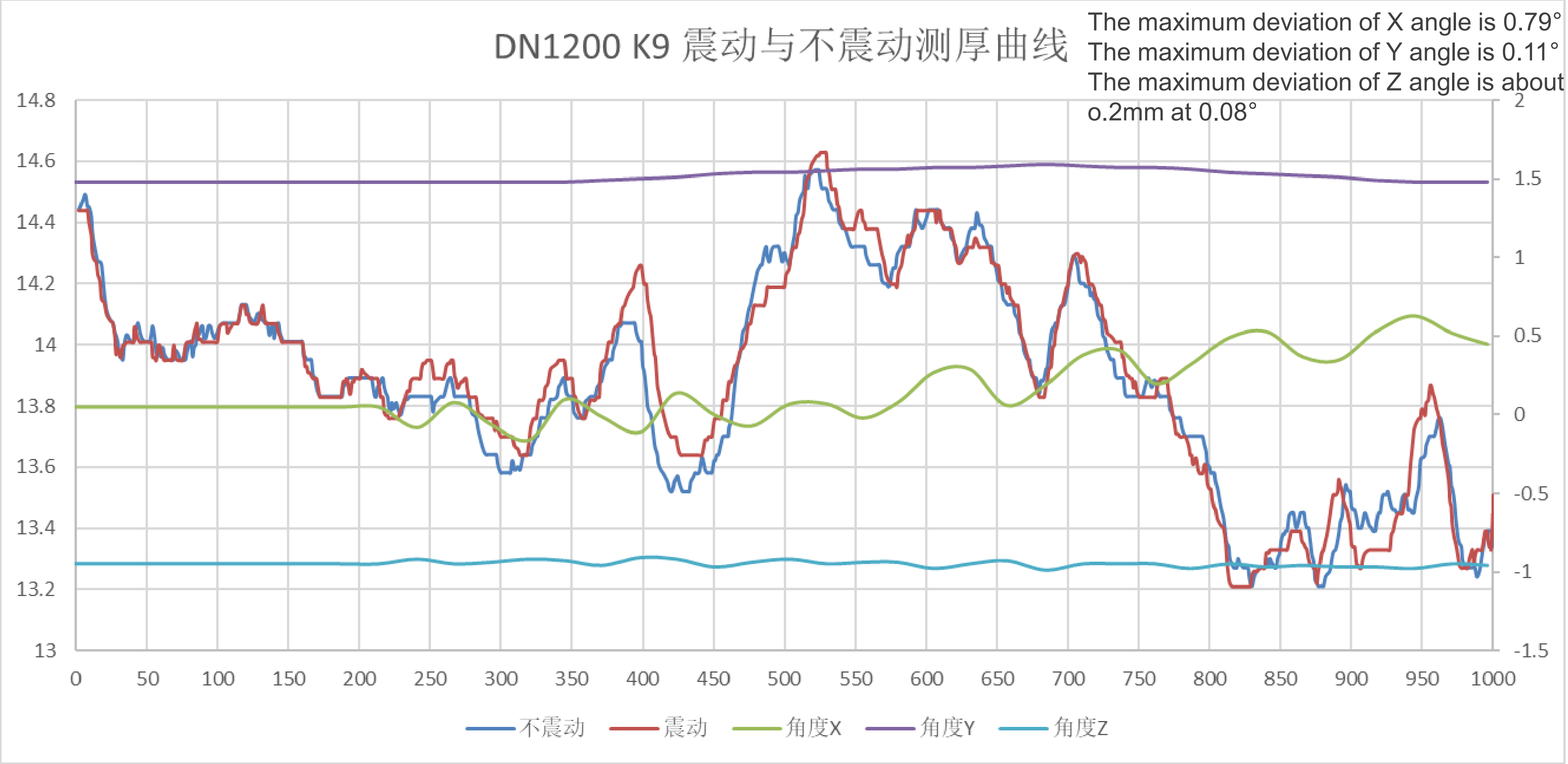
Vibration test - the measured object vibrates



The above figure shows the vibration of the segment under test, as can be seen in the figure, the effect is very small, below 0.1mm, the impact can be ignored

INTERMEDIATE INDUSTRY TEST

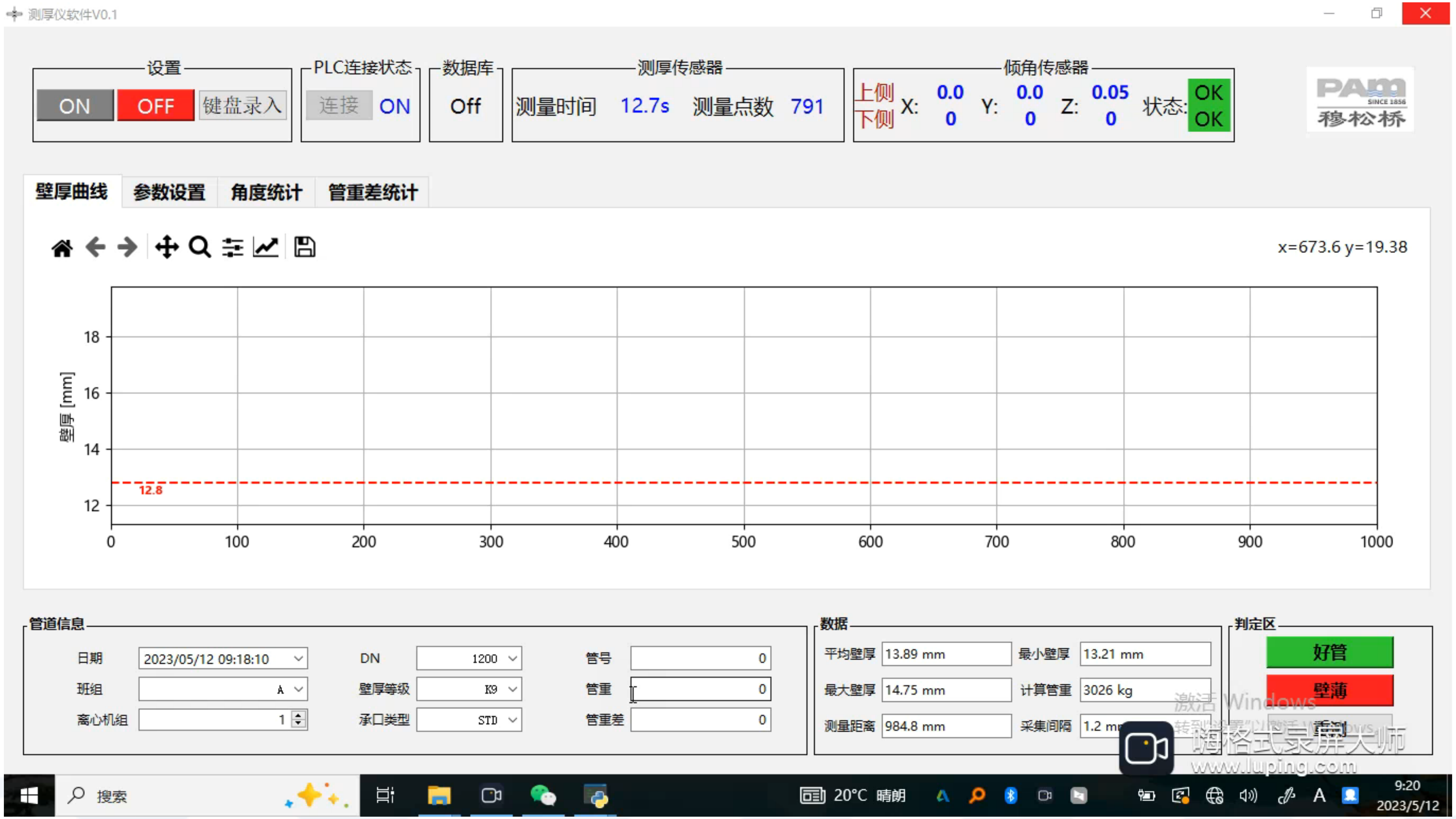
Thickness measurement curve and angle curve during vibration

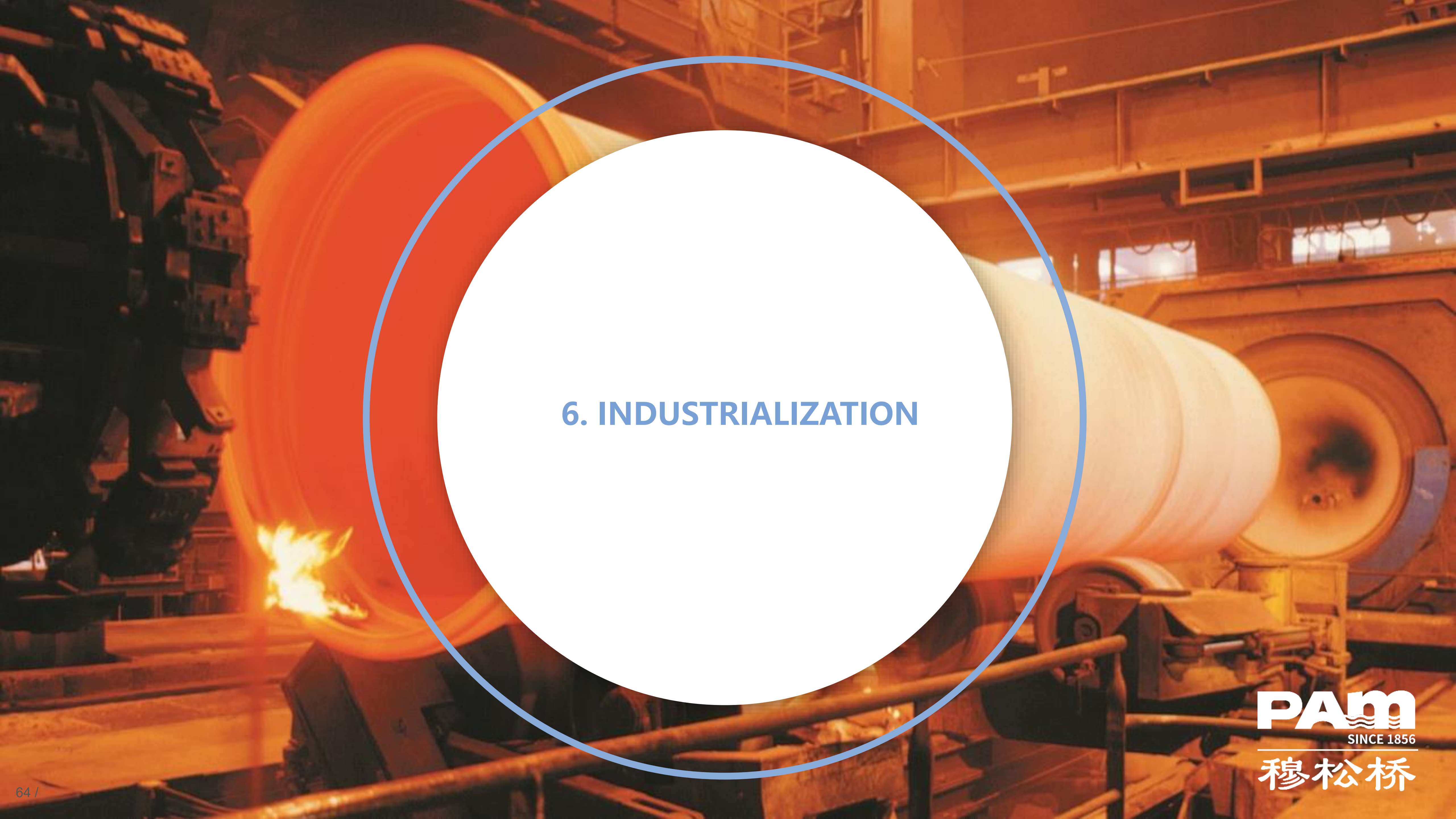


The vibration is increasing, but the wall thickness curve has no tendency to deviate more and more

INTERMEDIATE INDUSTRY TEST

Simulating vibration, see how angle changes

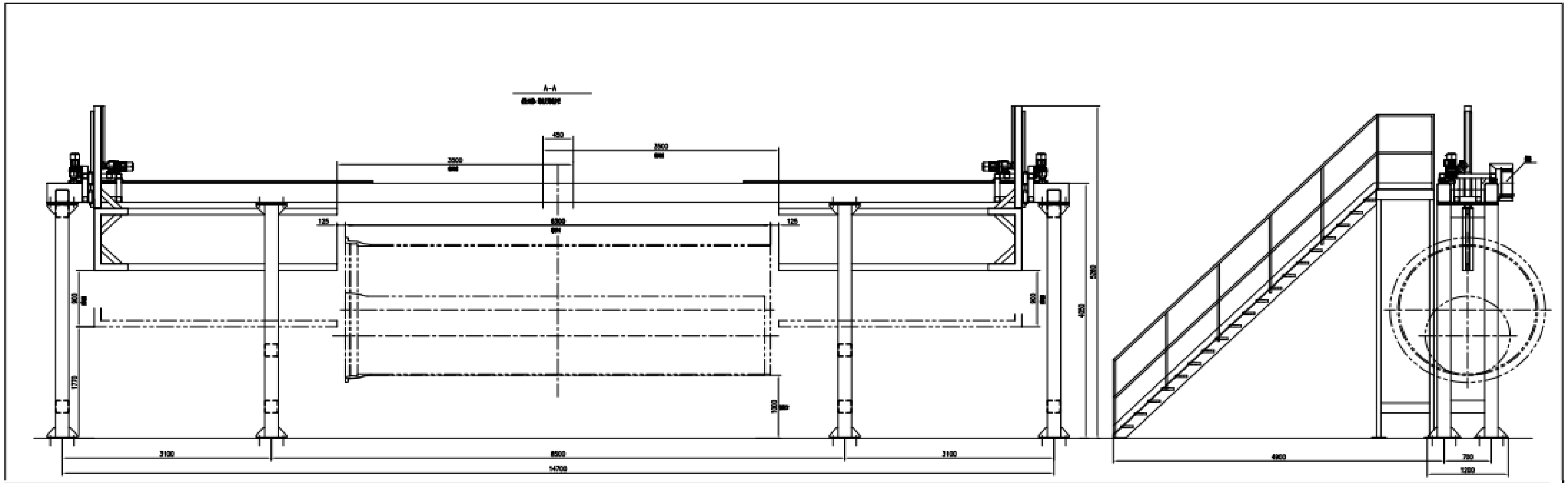




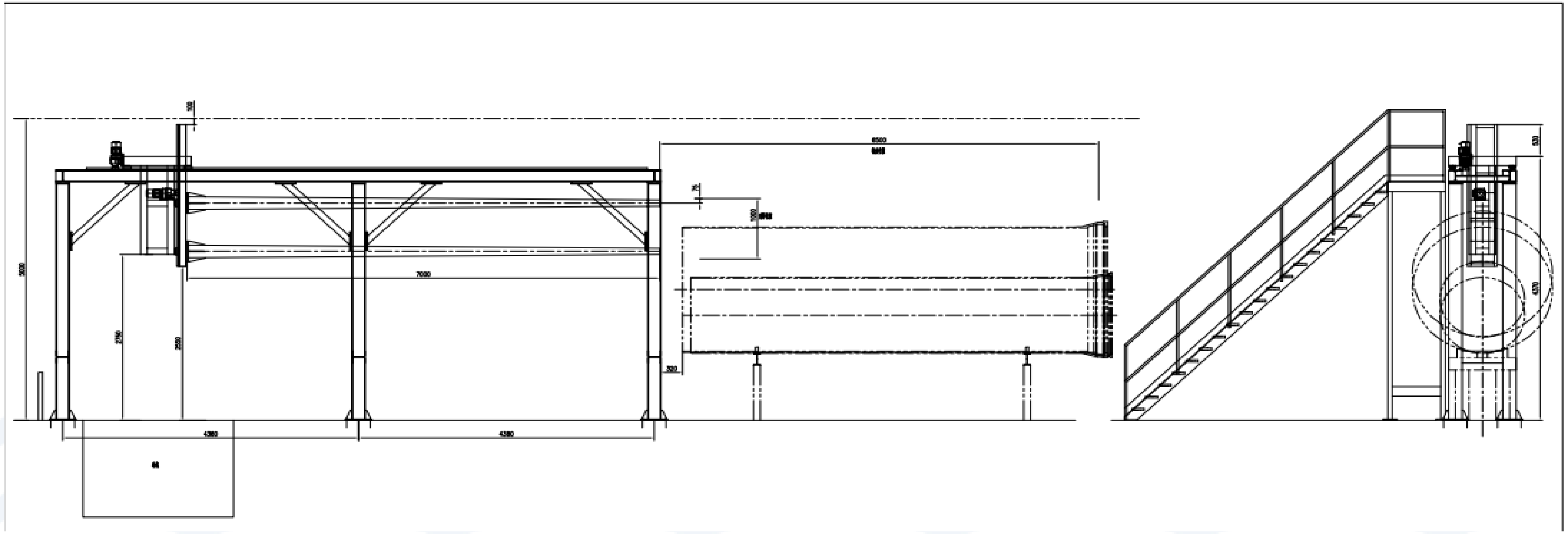
6. INDUSTRIALIZATION

INDUSTRIALIZATION

Equipment design program



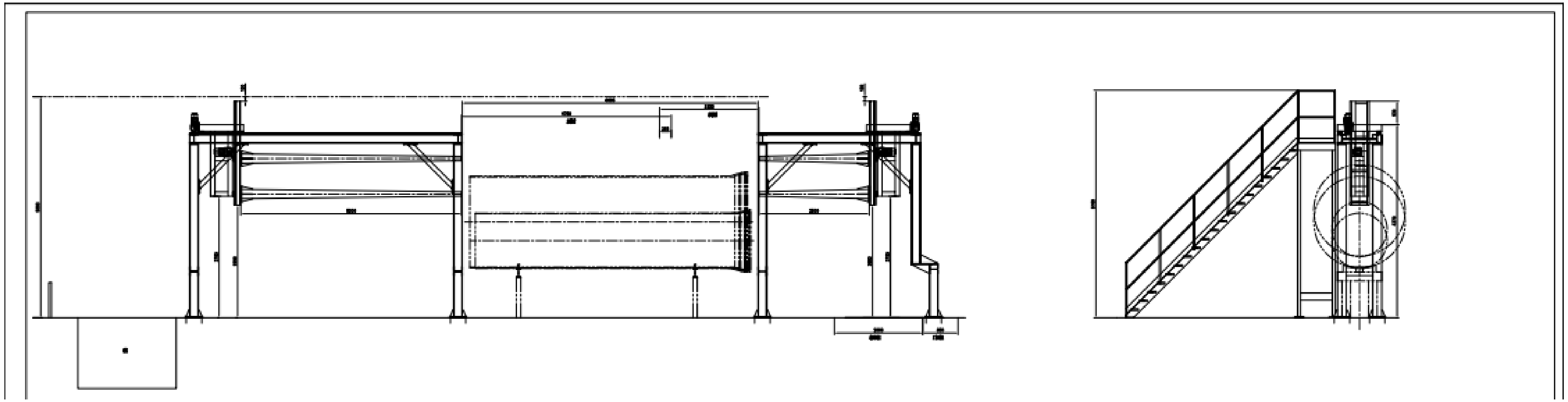
Design program 1



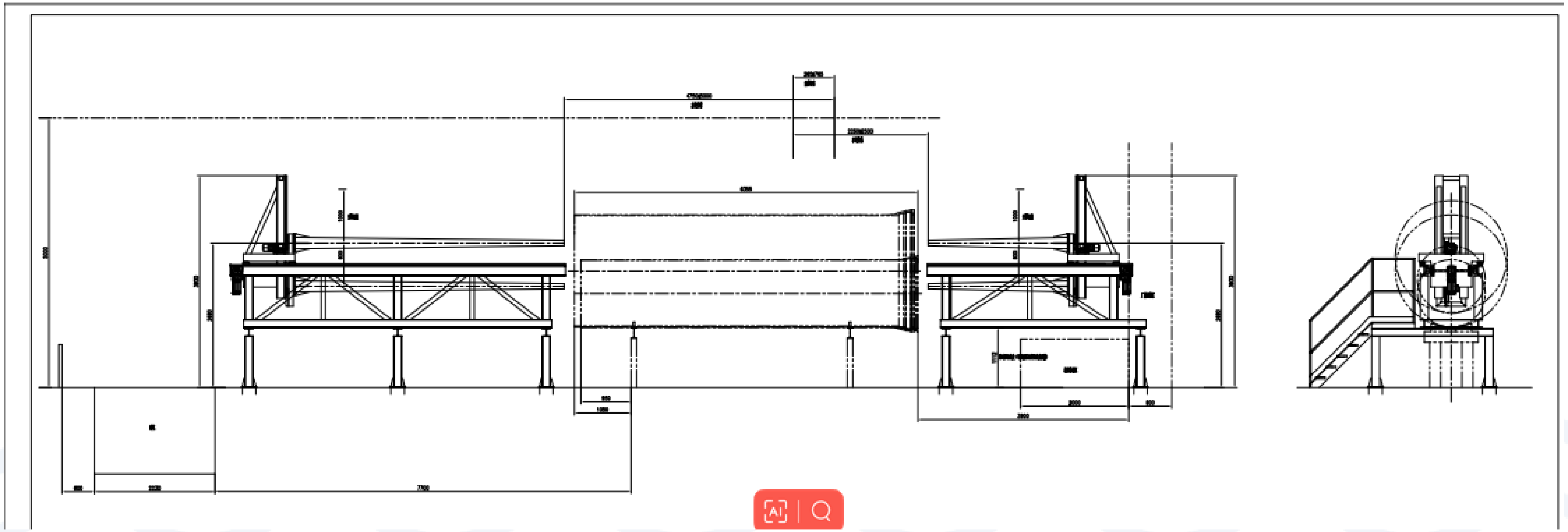
Design program 2

INDUSTRIALIZATION

Equipment design program



Design program 3



Design program 4

Optimization and final selection

Optimization

- The optimal solution is design program 1, the design of 3m on each side of the extension arm greatly reduces the accuracy error caused by the vibration of the extension arm.
- This design requires the least cost.
- However, due to the site constraints, this design cannot be applied in actual production.

Final selection

- In actual application, we finally choose design program 2. Design program 3 and 4 still constrained by the site.
- This design meets the site requirements and has less impact on accuracy

THANKS FOR YOUR ATTENTION!

PAM
SINCE 1856
穆松桥